

Exact two-colour Rado numbers of $x_1 + ax_2 - x_3 = c$: the window $0 < c < a$ and two conjectures of Dwivedi and Tripathi

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Abstract

For integers $a \geq 1$ and c , the 2-colour Rado number $\text{Rad}_2(a; c)$ is the least N such that every 2-colouring of $\{1, \dots, N\}$ admits a monochromatic solution of $x_1 + ax_2 - x_3 = c$, repeated values being allowed; it exists if and only if $\nu_2(a) \leq \nu_2(c)$, where ν_2 is the 2-adic valuation. Dwivedi and Tripathi [*Integers* **20** (2020), #A36] determined $\text{Rad}_2(a; c)$ in several parameter ranges and asked for the remaining exact values. We resolve the window $0 < c < a$ completely: writing $d = a - c$ and $r \equiv a \pmod{2d}$ with $r \in [1, 2d]$,

$$\text{Rad}_2(a; a - d) = \begin{cases} 2a + 2, & d = 1, \\ (a + 3)d + a - \min(r - 1, d - 1), & d \geq 2, \end{cases}$$

whenever the number exists. The upper-bound argument extends to $c \leq 0$, proving Conjecture 1 of Dwivedi and Tripathi in full; their Conjecture 2 then follows by a reflection argument. In every regime the final obstruction is precisely the 2-adic existence condition. In the remaining band $a < c < a(K + 2)$, where $K + 1 = \lceil (1 + c(a + 3)) / (1 + a(a + 3)) \rceil$ is the general lower bound of Dwivedi and Tripathi, the values appear to follow no closed formula; there we prove an extension of their multiples theorem and announce the complete determination of the band, with proofs to appear in a companion manuscript. All results are corroborated by more than 30,000 SAT-computed values with independently verifiable certificates.

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1 Introduction

For an equation $\mathcal{E}$ in the variables $x_1, \dots, x_m$, the 2-colour Rado number $\text{Rad}_2(\mathcal{E})$ is the least positive integer N such that every 2-colouring of the integer interval $[1, N]$ contains a monochromatic solution of $\mathcal{E}$, that is, a solution all of whose entries receive the same colour; solutions with repeated values are allowed. Following Dwivedi and Tripathi [1], we study the family

$$\mathcal{E}_{a,c}: \quad x_1 + ax_2 - x_3 = c,$$

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where a is a positive integer and c any integer, and write $\text{Rad}_2(a; c) := \text{Rad}_2(\mathcal{E}_{a,c})$.

Dwivedi and Tripathi proved that $\text{Rad}_2(a; c)$ exists if and only if $\nu_2(a) \leq \nu_2(c)$ (with $\nu_2(0) = \infty$), and determined it exactly in several parameter ranges: for $c \leq 0$ when $a \mid c$, or when a is odd and c is sufficiently negative (in a precise residue-dependent sense); for $c = ma$ with $1 < m \leq a + 1$; and for $c > a$ sufficiently large. In the remaining ranges they gave upper and lower bounds, together with two conjectures for the zones adjacent to their theorems. In the recent paper [2] the same authors returned to the problem and wrote that “it would be desirable to determine the exact value of the Rado number in all cases.”

In the window $0 < c < a$ their results give the bounds $(a + 3)(a - c) + 1 \leq \text{Rad}_2(a; c) \leq (2a + 1)(a - c) + 1$ (for odd a ; similarly for even a), and no conjecture. A subsequent paper of Arora, Dwivedi and Tripathi [3] extends the lower bound to a general class of equations $\sum_i a_i x_i - x_m = c$, with matching exact values when the coefficient multiset is 2-distributable; for the equation studied here the multiset is $\{1, a\}$, which is 2-distributable only for $a \leq 2$, so in our window that paper again yields exactly the lower bound $(a + 3)(a - c) + 1$. This note closes the window completely — in particular, the lower bound of [1, 3] is never tight there for $a \geq 3$, $d \geq 2$.

Theorem 1. *Let $a \geq 3$ be odd. Then $\text{Rad}_2(a; a - 1) = 2a + 2$.*

(For even a the number $\text{Rad}_2(a; a - 1)$ does not exist, since $a - 1$ is odd; so Theorem 1 covers the whole diagonal $c = a - 1$. The case $a = 3$, i.e. $\text{Rad}_2(3; 2) = 8$, appears in [1]; the general statement is new.)

Theorem 2. *Let $a > d \geq 2$ with $\nu_2(a) \leq \nu_2(a - d)$, and write $a = 2dq + r$ with $q \geq 0$ an integer and $r \in [1, 2d]$. Then*

$$\text{Rad}_2(a; a - d) = (a + 3)d + a - \min(r - 1, d - 1).$$

Under the existence condition one automatically has $r \in [1, 2d - 1]$ (Lemma 2.3), and the two theorems together determine $\text{Rad}_2(a; c)$ for every pair with $0 < c < a$ for which it exists. It is curious that the uniform formula of Theorem 2 fails for $d = 1$ by exactly one unit: the extremal colouring family of Section 4 degenerates at $d = 1$ into a colouring of $[1, 2a + 2]$ which is *not* valid — and the proof of Theorem 1 shows that nothing else works at that length either.

Our third and fourth results settle the two conjectures of [1]. For $c \leq 0$ one has $d = a - c \geq a$, so that $r \equiv a \pmod{2d}$ is simply $r = a$: the window formula of Theorem 2 formally reduces to $(a + 3)d + a - (a - 1) = (a + 3)(a - c) + 1$, which is exactly the value predicted by Conjecture 1 of [1]. This is no coincidence: our upper-bound proof for the regime $r \leq d$ extends to this degenerate case ($q = 0$ in the normalisation $a = 2dq + r$) after adjusting two range conditions, and the lower bound is Theorem 2 of [1].

Theorem 6 (Conjecture 1 of [1]). *Let $a \geq 1$ and $c \leq 0$ with $\nu_2(a) \leq \nu_2(c)$. Then $\text{Rad}_2(a; c) = (a + 3)(a - c) + 1$.*

Theorem 7 (Conjecture 2 of [1]). *Let $a \geq 1$ and let $c \geq a(K + 2)$ with $\nu_2(a) \leq \nu_2(c)$, where $K = \lceil (1 + c(a + 3)) / (1 + a(a + 3)) \rceil - 1$. Then $\text{Rad}_2(a; c) = K + 1$.*

Theorem 7 follows from Theorem 6 by a reflection argument generalising the proof of Theorem 7 of [1] (Section 7). Taken together, these results determine $\text{Rad}_2(a; c)$ exactly for every $c \leq 0$, every $0 < c \leq a$, and every $c \geq a(K + 2)$. The remaining zone — the band $a < c < a(K + 2)$ — behaves very differently: no closed formula holds there, the fluctuations being of Sturmian

type. Section 8 states what we can prove about the band: an extension of the multiples theorem of [1] (Theorem 8, proved here), the complete determination of the band’s deep tail, where the threshold is itself a Rado number from the window $0 < c < a$ (Theorem 9), and the exact determination of the whole band by a polynomial-time algorithm (Theorem 10); the proofs of the last two will appear in the companion manuscript [5].

Table 1 and Figure 1 summarise the resulting picture.

zone	$\text{Rad}_2(a; c)$	reference
$c \leq 0$	$(a + 3)(a - c) + 1$	Theorem 6
$0 < c < a, d = 1$	$2a + 2$	Theorem 1
$0 < c < a, d \geq 2$	$(a + 3)d + a - \min(r - 1, d - 1)$	Theorem 2
$c = a$	1	[1]
$a < c < a(K + 2), a \mid c$	c/a	Theorem 8
$a < c < a(K + 2), a \nmid c$	deep-tail formula; exact algorithm	Theorems 9–10*
$c \geq a(K + 2)$	$K + 1$	Theorem 7

Table 1: The value of $\text{Rad}_2(a; c)$ on each parameter zone, assuming the existence condition $\nu_2(a) \leq \nu_2(c)$. Here $d = a - c$, $r \equiv a \pmod{2d}$ with $r \in [1, 2d]$, and $K = \lceil (1 + c(a + 3)) / (1 + a(a + 3)) \rceil - 1$. Entries marked * are announced here and proved in the companion manuscript; all other entries are proved in this note or in [1].

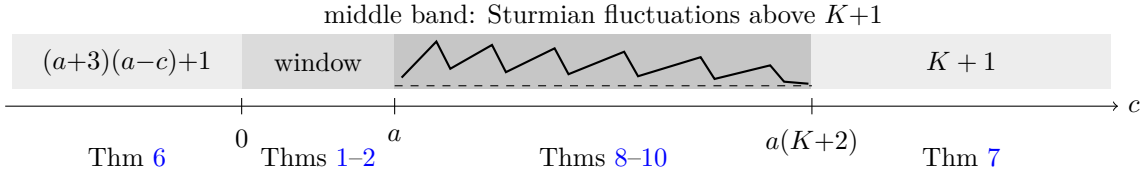


Figure 1: The zones of $\text{Rad}_2(a; c)$ (schematic, c increasing to the right). Outside the middle band the value follows the closed formulas of Table 1; inside it, the value oscillates above the lower bound $K + 1$ of [1] (dashed) in a quasi-periodic, three-distance pattern, and is determined by the algorithm of Theorem 10.

The proofs have a common architecture, which generalises the forcing arguments of [1]. The lower bounds are witnessed by an explicit two-parameter family of periodic colourings with regularly spaced phase defects (Section 4). The upper bounds are rigidity arguments: any putative valid colouring of the critical interval is forced, triple by triple, into a highly structured pattern, and the final obstruction that kills the pattern is, in both parameter regimes, *exactly* the 2-adic existence condition $\nu_2(a) \leq \nu_2(a - d)$ — in the regime $r \leq d$ through the fact that $\gcd(r, 2d) \mid d$, and in the regime $r > d$ through the incompatibility of antipodal and s -antiperiodic patterns on $\mathbb{Z}_{2d}$ when $\nu_2(s) \neq \nu_2(d)$, $s = r - d$ (Sections 5 and 6).

Methodology and verifiability. The formula of Theorem 2, the extremal family, and the key structural lemmas were first *discovered* through an extensive SAT-solver-based computation of exact Rado numbers: 3662 pairs (a, c) with $3 \leq a \leq 15$, $-100 \leq c \leq 300$, later extended along $0 < c < a$ up to $a = 45$. Every computed value $\text{Rad}_2(a; c) = N$ is certified in both directions: a valid colouring of $[1, N - 1]$ (checkable by a trivial independent program) and a DRAT proof of unsatisfiability at N , generated by CaDiCaL 2.1.3 [4] and verified by `drat-trim` against an

independently regenerated CNF encoding. The proofs presented here are, however, entirely self-contained and human-readable; the computations play no logical role in the theorems of this note: they served to discover the statements and to corroborate them independently. Every one of the 925 computed values with $c \leq 0$, and every value in the ranges of the four theorems, matches the formulas proved here (Section 9).

2 Preliminaries

Throughout, $[u, v]$ denotes $\{u, u+1, \dots, v\}$ and ν_2 the 2-adic valuation. A *colouring* is a map $\chi: [1, N] \rightarrow \{0, 1\}$; it is *valid* (for the pair (a, c)) if no solution $(x_1, x_2, x_3) \in [1, N]^3$ of $x_1 + ax_2 - x_3 = c$ is monochromatic.

Solving for x_3 , we see that the solutions in $[1, N]^3$ are exactly the triples

$$T(x, k) = (x, k, x + a(k-1) + d), \quad d := a - c, \quad (1)$$

with $x, k \in [1, N]$ and $1 \leq x + a(k-1) + d \leq N$ (for $d \geq 1$, which holds everywhere outside Section 8, the constraint $x + a(k-1) + d \geq 1$ is automatic). We record two standard facts.

Lemma 2.1 (monotonicity). *If $[1, N]$ admits a valid colouring, so does $[1, N']$ for every $N' \leq N$ (restrict the colouring). Hence $\text{Rad}_2(a; c) > N$ if and only if $[1, N]$ admits a valid colouring, and to determine $\text{Rad}_2(a; c) = N$ it suffices to exhibit a valid colouring of $[1, N-1]$ and to prove that no valid colouring of $[1, N]$ exists.*

Lemma 2.2 (colour symmetry). *χ is valid if and only if $1 - \chi$ is valid. In any proof by contradiction we may therefore normalise $\chi(1) = 0$.*

Lemma 2.3 (normalisation). *Let $d \geq 1$, $a > d$, and write $a = 2dq + r$ with $r \in [1, 2d]$.*

- (i) *If $r = 2d$, then $\nu_2(a) > \nu_2(a-d)$, so $\text{Rad}_2(a; a-d)$ does not exist. Under the existence condition, therefore, $r \in [1, 2d-1]$.*
- (ii) *If $r \leq d$ then $q \geq 1$ (so $a \geq 2d+1$).*
- (iii) *If $r = d + s$ with $s \in [1, d-1]$, then $q \geq 0$, and the existence condition is equivalent to $\nu_2(s) \neq \nu_2(d)$ (Lemma 6.7 below); if $r \leq d$, the existence condition implies $\nu_2(r) \leq \nu_2(d)$, hence $\gcd(r, 2d) \mid d$ (Lemma 5.1 below).*

Proof. (i) Write $a = 2dm$, $m \geq 1$. Then $\nu_2(a) \geq \nu_2(d) + 1$ while $a - d = d(2m-1)$ gives $\nu_2(a-d) = \nu_2(d)$. (ii) If $q = 0$ then $a = r \leq d$, contradicting $a > d$. (iii) The two 2-adic statements are proved as Lemmas 5.1 and 6.7 in the sections where they are used. $\square$

3 The diagonal $c = a - 1$: proof of Theorem 1

Here $d = 1$ and the solution triples (1) are $T(x, k) = (x, k, x + a(k-1) + 1)$.

Lower bound

Colour $[1, 2a + 1]$ by the classes

$$A = \{x \text{ odd} : x \leq a\} \cup \{x \text{ even} : a + 3 \leq x \leq 2a\}, \quad B = [1, 2a + 1] \setminus A.$$

If a triple $T(x, k)$ lies in $[1, 2a + 1]^3$ then $a(k - 1) \leq 2a - 1$, so $k \in \{1, 2\}$. A monochromatic triple in A would need $k = 1$ (as $2 \notin A$: the even elements of A start at $a + 3$); but for $x \in A$ odd, $x + 1$ is even and $\leq a + 1$, hence in B , and for $x \in A$ even, $x + 1$ is odd and $\geq a + 4$, hence in B . A monochromatic triple in B would need $k = 2$ (as $1 \notin B$); but for $x \in B$ even, $x \leq a + 1$: if $x \leq a - 1$ then $x_3 = x + a + 1$ is even in $[a + 3, 2a]$, hence in A , and if $x = a + 1$ then $x_3 = 2a + 2$ is out of range; for $x \in B$ odd, $x \geq a + 2$, so $x_3 \geq 2a + 3$ is out of range. Hence the colouring is valid and $\text{Rad}_2(a; a - 1) \geq 2a + 2$.

Upper bound

Let χ colour $[1, 2a + 2]$ with no monochromatic triple; by Lemma 2.2, $\chi(1) = 0$.

Step 1. $T(x, 1) = (x, 1, x + 1)$ is a solution for every $x \leq 2a + 1$. Since $\chi(1) = 0$, no two consecutive integers are both coloured 0. In particular $T(1, 1) = (1, 1, 2)$ forces $\chi(2) = 1$.

Step 2. $T(x, 2) = (x, 2, x + a + 1)$ is a solution for $x \leq a + 1$. Since $\chi(2) = 1$: if $\chi(x) = 1$ with $x \leq a + 1$, then $\chi(x + a + 1) = 0$.

Step 3. The pairs $\{x, x + a + 1\}$, $x \in [1, a + 1]$, partition $[1, 2a + 2]$. By Step 2 each pair contains a 0, so $\#\chi^{-1}(0) \geq a + 1$; by Step 1, $\#\chi^{-1}(0) \leq a + 1$. Hence exactly $a + 1$ zeros, no two adjacent, in an interval of length $2a + 2$: the zero set must be

$$Z_j = \{1, 3, \dots, 2j - 3\} \cup \{2j, 2j + 2, \dots, 2a + 2\}$$

for some j , and $\chi(1) = 0$ forces $j \in [2, a + 2]$.

Step 4. Each j is contradictory. If $j = a + 2$ (zeros are all the odds), the triple $(a + 1, 2, 2a + 2)$ is monochromatic of colour 1 ($a + 1$ and $2a + 2$ are even). If $3 \leq j \leq a + 1$, then $\chi(1) = \chi(3) = 0$ and $2a + 2 \geq 2j$ is an even member of Z_j ; the triple $(1, 3, 2a + 2)$ is monochromatic of colour 0. If $j = 2$ (zeros are $\{1\} \cup \{4, 6, \dots, 2a + 2\}$), then $\chi(3) = \chi(2) = 1$ and $a + 4$ is odd and ≥ 5 , hence coloured 1; the triple $(3, 2, a + 4)$ is monochromatic of colour 1 (note $a + 4 \leq 2a + 2$ for $a \geq 2$). $\square$

4 The extremal family and the lower bound for $d \geq 2$

Throughout this section fix $d \geq 2$ and $a \geq d + 1$ with $\nu_2(a) \leq \nu_2(a - d)$, and write $a = 2dq + r$ with $q \geq 0$ and $r \in [1, 2d - 1]$, as provided by Lemma 2.3. We distinguish two regimes. In *regime 1* we have $r \leq d$; then $q \geq 1$, since $q = 0$ would give $a = r \leq d$, contradicting $a \geq d + 1$; in particular $a \geq 2d + 1$ in regime 1. In *regime 2* we have $r \geq d + 1$; we set $s = r - d \in [1, d - 1]$, and here $q \geq 0$, the value $q = 0$ occurring precisely when $d + 1 \leq a \leq 2d - 1$. Throughout we set

$$t = \max(d, 2d - r), \quad N^* = (a + 3)d + a - \min(r - 1, d - 1) - 1,$$

so that $t = 2d - r \in [d, 2d - 1]$ and $N^* = (a + 3)d + a - r$ in regime 1, while $t = d$ and $N^* = (a + 2)d + a$ in regime 2 (as $\min(r - 1, d - 1)$ equals $r - 1$ for $r \leq d$ and $d - 1$ for $r \geq d + 1$). Recall from (1) that the solutions of $x_1 + ax_2 - x_3 = a - d$ with all values in $[1, N]$ are exactly the triples (x, k, y) with $y = x + a(k - 1) + d$ and $x, k, y \in [1, N]$. We shall exhibit a 2-colouring χ

of $[1, N^*]$ admitting no monochromatic such triple. The colouring is defined by an explicit run-length list (Lemma 4.1) and then rewritten in a *chunk-offset normal form* $\chi(x) = \alpha(w(x) - 1)$ (Lemma 4.2); for every in-range triple the offset increment obeys a trichotomy (Lemma 4.4), forcing in particular $k \leq d + 2$, and each of the three resulting cases leads to a colour conflict (Theorem 3). The analysis is uniform in $q \geq 1$ (regime 1), respectively $q \geq 0$ (regime 2), and covers the edge values $r = 1$, $r = d$, $r = d + 1$, $r = 2d - 1$ and $a = d + 1$.

Lemma 4.1 (the extremal family colouring). *Keep the notation above, with $r \in [1, 2d - 1]$. Let χ be the colouring of an initial interval of the positive integers obtained by concatenating runs of constant colour, the colours alternating and the first run having colour 1 (so the run of index j has colour 1 if j is odd and colour 0 if j is even), with run-length sequence*

$$\begin{aligned} \text{regime 1 } (r \leq d): & \quad [d]^{2q+1} ([d+r] [d]^{2q-1})^{d-1} [d+r] [d]^{2q+1}; \\ \text{regime 2 } (r > d, s = r - d): & \quad [d]^{2q+2} [s] ([d]^{2q+1} [s])^{d-1} [d]^{2q+2} [s], \end{aligned}$$

where $[\ell]^m$ abbreviates m consecutive runs of length ℓ and $[\ell] = [\ell]^1$. Then: (a) every run length is at least 1; (b) the total length equals N^* , so that $\chi: [1, N^*] \rightarrow \{0, 1\}$; (c) in regime 1 the i -th long $(d+r)$ -run has run index $2qi + 2$, which is even, hence colour 0; in regime 2 every s -run except the last has odd index, hence colour 1, while the final s -run has even index, hence colour 0.

Proof. (a) In regime 1: $d \geq 2$, $d+r \geq 3$, and $2q-1 \geq 1$ because $q \geq 1$; the multiplicities $2q+1$, $2q-1$ are positive. In regime 2: $d \geq 2$, $s \in [1, d-1]$, and $2q+1 \geq 1$, $2q+2 \geq 2$ for $q \geq 0$.

(b) Regime 1: using $2dq = a - r$, the total length is

$$2d(2q+1) + d(d+r) + (d-1)d(2q-1) = (4dq+2d) + (d^2+dr) + (2qd^2-d^2-2qd+d) = 2qd^2 + dr + 3d + 2dq,$$

while $N^* = (a+3)d + a - r = (2dq + r + 3)d + 2dq = 2qd^2 + rd + 3d + 2dq$; the two agree. Regime 2: using $2dq = a - d - s$ we have $d(2q+2) = a + d - s$ and $d(2q+1) = a - s$, so the total length is

$$2(a+d-s) + (d-1)(a-s) + (d+1)s = 2a + 2d - 2s + (d-1)a - (d-1)s + (d+1)s = (d+1)a + 2d,$$

since the coefficient of s is $-2 - (d-1) + (d+1) = 0$; and $(d+1)a + 2d = ad + a + 2d = (a+2)d + a = N^*$.

(c) Regime 1: the number of runs strictly preceding the i -th long run is $(2q+1) + (i-1)(1 + (2q-1)) = (2q+1) + 2q(i-1)$, so its index is $2qi + 2$, which is even; hence colour 0. Regime 2: the initial group $[d]^{2q+2} [s]$ contributes $2q+3$ runs, so the first s -run has index $2q+3$, odd: colour 1. Each subsequent middle group $[d]^{2q+1} [s]$ contributes $2q+2$ runs, so the s -run closing the i -th middle group ($1 \leq i \leq d-1$) has index $(2q+3) + i(2q+2)$, odd + even = odd: colour 1. In the final group, the first d -run has index $(2q+3) + (d-1)(2q+2) + 1$, which is even (odd + even + 1); its last d -run has index $(2q+3) + d(2q+2)$, odd; and the final s -run has index $(2q+3) + d(2q+2) + 1$, even: colour 0. $\square$

Lemma 4.2 (chunk-offset normal form). *Keep the notation of Lemma 4.1. Define the jump points $p_j = ja + t$ for $j = 1, \dots, d$, the jump count $J(x) = \#\{j \in [1, d] : x > p_j\}$, the offset $w(x) = x - aJ(x)$, and*

$$\alpha(u) = \begin{cases} 1 & \text{if } u \bmod 2d \in [0, d-1], \\ 0 & \text{if } u \bmod 2d \in [d, 2d-1]. \end{cases}$$

Then: (I) $\chi(x) = \alpha(w(x) - 1)$ for every $x \in [1, N^*]$. (II) (offset ranges) $J(x) = 0$ if and only if $x \in [1, a + t]$, and then $w(x) = x \in [1, a + t]$; for $j \in [1, d - 1]$, $J(x) = j$ if and only if $x \in [ja + t + 1, (j + 1)a + t]$, and then $w(x) \in [t + 1, a + t]$; and $J(x) = d$ if and only if $x \in [da + t + 1, N^*]$, and then $w(x) \in [t + 1, a + t + d]$. In particular $w(z) \geq 1$ for every $z \in [1, N^*]$; moreover $w(z) \leq a + t + d$ always, $w(z) \geq t + 1$ whenever $J(z) \geq 1$, and $w(z) \leq a + t$ whenever $J(z) \leq d - 1$.

Proof. We use two elementary facts. *Fact A:* $\alpha(u + d) = 1 - \alpha(u)$ for every integer u . Indeed, write $u = 2dm + v$ with $v \in [0, 2d)$: if $v < d$ then $(u + d) \bmod 2d = v + d \in [d, 2d)$, and if $v \geq d$ then $(u + d) \bmod 2d = v - d \in [0, d)$. *Fact B:* if $u \equiv 0 \pmod{2d}$ then $\alpha(u) = \alpha(u + 1) = \dots = \alpha(u + d - 1) = 1$, and if $u \equiv d \pmod{2d}$ then those d values all have $\alpha = 0$. Throughout the proof we use $a \equiv r \pmod{2d}$.

We first prove (II). We have $0 < p_1 < p_2 < \dots < p_d$ and, in both regimes, $N^* - p_d = N^* - da - t = a + d > 0$: in regime 1, $N^* - da - (2d - r) = (ad + a + 3d - r) - ad - 2d + r = a + d$; in regime 2, $N^* - da - d = (ad + a + 2d) - ad - d = a + d$. Hence the level sets $\{x \in [1, N^*] : J(x) = j\}$ are exactly the stated intervals, and $w(x) = x - ja$ on the j -th of them; subtracting ja from those intervals gives the stated offset ranges ($[ja + t + 1, (j + 1)a + t] - ja = [t + 1, a + t]$ and $[da + t + 1, N^*] - da = [t + 1, a + t + d]$). For the explicit lower bound: if $J(z) = 0$ then $w(z) = z \geq 1$, and if $J(z) \geq 1$ then $w(z) \geq t + 1 \geq 1$; so $w(z) \geq 1$ for every z . The remaining assertions are read off directly from the three ranges.

Proof of (I) in regime 1 ($t = 2d - r$, $q \geq 1$). By Lemma 4.1, $[1, N^*]$ decomposes into the consecutive segments

$$S_0 = [1, a + d - r], \quad L_i = [ia + d - r + 1, ia + 2d] \ (i = 1, \dots, d), \quad T_i = [ia + 2d + 1, (i + 1)a + d - r] \ (i \leq d - 1),$$

and $T_d = [da + 2d + 1, N^*]$. Here S_0 consists of the initial $2q + 1$ d -runs, of total length $d(2q + 1) = 2dq + d = a + d - r$; each L_i is the i -th long run, of length $d + r$; each T_i with $i \leq d - 1$ has length $a - d - r = d(2q - 1)$, and T_d has length $N^* - da - 2d = a + d - r = d(2q + 1)$. Consecutiveness: the segment before L_1 ends at $a + d - r$, and $\text{end}(S_0) + 1 = a + d - r + 1 = \text{start}(L_1)$, $\text{end}(L_i) + 1 = ia + 2d + 1 = \text{start}(T_i)$, $\text{end}(T_i) + 1 = (i + 1)a + d - r + 1 = \text{start}(L_{i+1})$, and $\text{end}(T_d) = N^*$. So the segments partition $[1, N^*]$.

Jump counts per segment. Here $p_j = ja + 2d - r$. On S_0 : $x \leq a + d - r < a + 2d - r = p_1$, so $J = 0$. Each L_i splits at p_i ; note $ia + d - r + 1 \leq p_i \leq ia + 2d - 1$ since $d \geq 1$ and $r \geq 1$. On $L_i^- = [ia + d - r + 1, p_i]$ we have $p_{i-1} = (i - 1)a + 2d - r < ia + d - r + 1$ if and only if $2d < a + d + 1$, i.e. $d \leq a$, which is true; so $J = i - 1$ on L_i^- . On $L_i^+ = [p_i + 1, ia + 2d]$ we have $J = i$ (for $i < d$ this uses $ia + 2d \leq p_{i+1} = ia + a + 2d - r$, i.e. $r \leq a$, true). On T_i : $p_i < ia + 2d + 1$ and, for $i < d$, $(i + 1)a + d - r < p_{i+1}$, so $J = i$; on T_d , $J = d$.

Phases. Write $u(x) = w(x) - 1$. On S_0 : $u = x - 1$ ranges over $[0, d(2q + 1) - 1]$, and S_0 consists of the d -runs $R_m = [(m - 1)d + 1, md]$, $m = 1, \dots, 2q + 1$, coloured 1 for m odd and 0 for m even (their global run indices are $1, \dots, 2q + 1$). For $x \in R_m$, $u = (m - 1)d + v$ with $v \in [0, d - 1]$; if m is odd then $(m - 1)d \equiv 0 \pmod{2d}$ and Fact B gives $\alpha = 1$; if m is even then $(m - 1)d \equiv d$ and $\alpha = 0$. This matches χ . On L_i^- : $u = x - 1 - (i - 1)a \in [a + d - r, a + 2d - r - 1]$, a set of d consecutive integers the first of which is $\equiv r + d - r = d \pmod{2d}$; by Fact B, $\alpha = 0$ on all of L_i^- . On L_i^+ : $u = x - 1 - ia \in [2d - r, 2d - 1]$, a subset of $[d, 2d - 1]$ because $r \leq d$; so $\alpha = 0$. Hence $\alpha = 0$ on the whole long run, matching its colour 0 (Lemma 4.1(c)). On T_i : $u = x - 1 - ia \in [2d, 2d + M - 1]$ with $M = d(2q - 1)$ for $i < d$ and $M = d(2q + 1)$ for $i = d$; thus $u - 2d$ starts at 0, and T_i consists of consecutive d -runs whose first has global index $2qi + 3$ (odd, colour 1: it is the run following the i -th long run, whose index is $2qi + 2$ by Lemma 4.1(c)).

Exactly as on S_0 , the d -blocks are aligned ($u \equiv 0 \pmod{2d}$ at the start of T_i) and α matches the alternating colours on all of T_i .

Proof of (I) in regime 2 ($t = d$, $s = r - d \in [1, d - 1]$, $q \geq 0$). Segments:

$$C_0^b = [1, a + d - s], \quad C_0^s = [a + d - s + 1, a + d],$$

and for $i = 1, \dots, d - 1$:

$$C_i^b = [ia + d + 1, (i + 1)a + d - s], \quad C_i^s = [(i + 1)a + d - s + 1, (i + 1)a + d],$$

and finally

$$F^b = [da + d + 1, da + a + 2d - s], \quad F^s = [da + a + 2d - s + 1, N^*].$$

Here C_0^b consists of the initial $2q + 2$ d -runs, of length $d(2q + 2) = (a - d - s) + 2d = a + d - s$; C_0^s is the first s -run; C_i^b consists of $2q + 1$ d -runs, of length $a - s = d(2q + 1)$; C_i^s is an s -run; F^b consists of $2q + 2$ d -runs, of length $a + d - s = d(2q + 2)$; and F^s is the final s -run (since $N^* = da + a + 2d$, its length is s). These segments are consecutive and partition $[1, N^*]$.

Jump counts. Here $p_j = ja + d$. On $C_0^b \cup C_0^s = [1, a + d] = [1, p_1]$: $J = 0$. On $C_i^b \cup C_i^s = [ia + d + 1, (i + 1)a + d] = [p_i + 1, p_{i+1}]$: $J = i$. On $F^b \cup F^s = [p_d + 1, N^*]$: $J = d$.

Phases ($u = w - 1$; note $a \equiv r = d + s \pmod{2d}$). On C_0^b : $u = x - 1 \in [0, d(2q + 2) - 1]$ with $d(2q + 2) = 2d(q + 1)$; the d -blocks are aligned starting at phase 0, so α alternates $1, 0, 1, \dots$ run by run, matching the global run indices $1, \dots, 2q + 2$ (odd index = colour 1). On C_0^s : $u = x - 1 \in [a + d - s, a + d - 1]$; since $a + d - s = 2d(q + 1) \equiv 0 \pmod{2d}$, we get $u \bmod 2d \in [0, s - 1]$, a subset of $[0, d - 1]$ (as $s \leq d - 1$): so $\alpha = 1$, matching colour 1 (index $2q + 3$, odd, by Lemma 4.1(c)). On C_i^b : $u = x - 1 - ia \in [d, d + d(2q + 1) - 1]$; the d -blocks are aligned starting at phase d , so α alternates $0, 1, 0, \dots$, matching the colours, since the first run of C_i^b has even global index $(2q + 3) + (i - 1)(2q + 2) + 1$ (odd + even + 1). On C_i^s : $u = x - 1 - ia \in [d + a - s, d + a - 1]$; since $d + a - s \equiv d + (d + s) - s = 2d \equiv 0 \pmod{2d}$, we get $u \bmod 2d \in [0, s - 1]$: so $\alpha = 1$, matching colour 1 (index $(2q + 3) + i(2q + 2)$, odd). On F^b : $u = x - 1 - da \in [d, d + 2d(q + 1) - 1]$; the d -blocks are aligned starting at phase d , so α alternates $0, 1, \dots, 1$ over the $2q + 2$ runs, ending with $\alpha = 1$, matching the colours (the first run has even index $(2q + 3) + (d - 1)(2q + 2) + 1$, colour 0; the last has odd index $(2q + 3) + d(2q + 2)$, colour 1). On F^s : $u = x - 1 - da \in [d + 2d(q + 1), d + 2d(q + 1) + s - 1]$, so $u \bmod 2d \in [d, d + s - 1]$, a subset of $[d, 2d - 1]$ (as $s \leq d - 1$): so $\alpha = 0$, matching the final s -run's colour 0 (index $(2q + 3) + d(2q + 2) + 1$, even). $\square$

Lemma 4.3 (colours of the small integers). *With χ as above ($d \geq 2$, $r \in [1, 2d - 1]$): $\chi(k) = 1$ for $1 \leq k \leq d$, and $\chi(k) = 0$ for $k \in \{d + 1, d + 2\}$.*

Proof. Since $a \geq d + 1$ and $t \geq d$, we have $a + t \geq 2d + 1 > d + 2$, the last inequality because $d \geq 2$ (indeed $2d + 1 > d + 2$ if and only if $d > 1$). Hence every $k \leq d + 2$ satisfies $k \leq a + t = p_1$, so $J(k) = 0$ and $w(k) = k$ (Lemma 4.2(II)); also $d + 2 < a + t \leq N^*$, so $\chi(k)$ is defined. By Lemma 4.2(I), $\chi(k) = \alpha(k - 1)$. For $1 \leq k \leq d$: $k - 1 \in [0, d - 1]$, so $\alpha = 1$. For $k \in \{d + 1, d + 2\}$: $k - 1 \in \{d, d + 1\}$, and $d + 1 \leq 2d - 1$ because $d \geq 2$, so $k - 1 \in [d, 2d - 1]$ and $\alpha = 0$. (This is the only place where the hypothesis $d \geq 2$ is used, and it is sharp: for $d = 1$ one has $\chi(d + 2) = \chi(3) = 1$, and the proof of Theorem 3 breaks precisely in Case $n = +1$, sub-case $J(y) = d$; see Remark 1.) $\square$

Lemma 4.4 (jump-count trichotomy). *Let $x, y \in [1, N^*]$ with $y = x + a(k - 1) + d$ for some integer $k \geq 1$. Put $\Delta J = J(y) - J(x)$ and $n = k - 1 - \Delta J$. Then $w(y) = w(x) + na + d$, $\Delta J \geq 0$, and $n \in \{-1, 0, +1\}$. Consequently $k = n + 1 + \Delta J \leq d + 2$ for every in-range triple.*

Proof. The identity: $w(y) - w(x) = (y - aJ(y)) - (x - aJ(x)) = (y - x) - a\Delta J = a(k - 1) + d - a\Delta J = na + d$. And $\Delta J \geq 0$ because $y > x$ (as $a(k - 1) + d > 0$) and J is non-decreasing.

Suppose $n \geq 2$. Then $w(y) = w(x) + na + d \geq 1 + 2a + d$, using $w(x) \geq 1$ (Lemma 4.2(II)). But $w(y) \leq a + t + d$ always (Lemma 4.2(II)), so $2a + d + 1 \leq a + t + d$, i.e. $a + 1 \leq t$. In regime 1, $a \geq 2d + 1$ while $t = 2d - r \leq 2d - 1$, so $a + 1 \geq 2d + 2 > t$: contradiction. In regime 2, $t = d < d + 2 \leq a + 1$: contradiction.

Suppose $n \leq -2$. Then $\Delta J = k - 1 - n \geq k + 1 \geq 2$. Hence $J(y) = J(x) + \Delta J \geq 2 \geq 1$, so $w(y) \geq t + 1$ (Lemma 4.2(II)); and $J(x) = J(y) - \Delta J \leq d - 2 \leq d - 1$, so $w(x) \leq a + t$ (Lemma 4.2(II)). Then

$$t + 1 \leq w(y) = w(x) + na + d \leq (a + t) - 2a + d = t - (a - d) \leq t - 1,$$

since $a \geq d + 1$: contradiction.

So $n \in \{-1, 0, 1\}$. Finally $\Delta J \leq J(y) \leq d$ gives $k = n + 1 + \Delta J \leq 1 + 1 + d = d + 2$. $\square$

Theorem 3 (lower bound for $d \geq 2$). *Let $d \geq 2$ and $a \geq d + 1$ with $\nu_2(a) \leq \nu_2(a - d)$, and let r, t, N^*, χ be as above. Then χ is a 2-colouring of $[1, N^*]$ containing no monochromatic triple (x, k, y) with $y = x + a(k - 1) + d$ and $x, k, y \in [1, N^*]$. Consequently*

$$\text{Rad}_2(a; a - d) \geq N^* + 1 = (a + 3)d + a - \min(r - 1, d - 1),$$

i.e. the value asserted in Theorem 2 is a lower bound for $\text{Rad}_2(a; a - d)$ for every such pair (a, d) .

Proof. By Lemma 2.3 we have $r \in [1, 2d - 1]$, so Lemmas 4.1–4.4 apply. Suppose, for contradiction, that x, k and $y = x + a(k - 1) + d$ all lie in $[1, N^*]$ and $\chi(x) = \chi(k) = \chi(y)$. Let $\Delta J = J(y) - J(x)$ and $n = k - 1 - \Delta J$; by Lemma 4.4, $n \in \{-1, 0, 1\}$ and $w(y) = w(x) + na + d$. Write α, w, J as in Lemma 4.2 and recall $\chi(z) = \alpha(w(z) - 1)$ for all $z \in [1, N^*]$; Facts A and B below refer to the proof of Lemma 4.2.

Case $n = 0$. Then $w(y) = w(x) + d$, so $\chi(y) = \alpha(w(x) - 1 + d) = 1 - \alpha(w(x) - 1) = 1 - \chi(x)$ by Fact A. Hence $\chi(x) \neq \chi(y)$: contradiction.

Case $n = -1$, i.e. $\Delta J = k$. Then $k = \Delta J \leq J(y) \leq d$, so $\chi(k) = 1$ by Lemma 4.3; the common colour is 1. Also $J(y) = J(x) + k \geq 1$, so $w(y) \geq t + 1$; and $J(x) = J(y) - k \leq d - 1$, so $w(x) \leq a + t$ (Lemma 4.2(II)). From $w(y) = w(x) - (a - d)$ we get $w(x) \geq w(y) + a - d \geq a + t + 1 - d$ and $w(y) \leq (a + t) - (a - d) = t + d$. Now: in regime 1 ($t = 2d - r$), $w(x) - 1$ lies in $[a + t - d, a + t - 1]$, a set of d consecutive integers the first of which satisfies $a + t - d = a + d - r \equiv r + d - r = d \pmod{2d}$; by Fact B, $\alpha(w(x) - 1) = 0$, i.e. $\chi(x) = 0 \neq 1$: contradiction. In regime 2 ($t = d$), $w(y) - 1$ lies in $[t, t + d - 1] = [d, 2d - 1]$, already inside $[0, 2d)$, so $\alpha(w(y) - 1) = 0$, i.e. $\chi(y) = 0 \neq 1$: contradiction.

Case $n = +1$, i.e. $\Delta J = k - 2$. Since $\Delta J \geq 0$, we have $k \geq 2$. Here $w(y) = w(x) + a + d$. Since $w(y) \leq a + t + d$ always (Lemma 4.2(II)), $w(x) \leq t$; but every z with $J(z) \geq 1$ has $w(z) \geq t + 1$, so $J(x) = 0$ and $x = w(x) \in [1, t]$. Hence $\Delta J = J(y) = k - 2$.

Sub-case $J(y) \leq d - 1$. Then $w(y) \leq a + t$ (Lemma 4.2(II)), so $x = w(y) - a - d \leq t - d$. In regime 2, $t = d$ gives $x \leq 0$: impossible. In regime 1, $x \in [1, t - d] = [1, d - r]$ (nonempty only if $r \leq d - 1$). Then $\chi(x) = \alpha(x - 1)$ with $x - 1 \in [0, d - r - 1]$, a subset of $[0, d - 1]$: $\chi(x) = 1$.

And $w(y) - 1 = x + a + d - 1 \equiv (x - 1) + r + d \pmod{2d}$, with $(x - 1) + r + d \in [r + d, 2d - 1]$, a subset of $[d, 2d - 1]$: $\chi(y) = 0$. So $\chi(x) \neq \chi(y)$: contradiction.

Sub-case $J(y) = d$. Then $k - 2 = d$, so $k = d + 2$ and $\chi(k) = 0$ by Lemma 4.3 (here $d + 2 \leq 2d$, i.e. $d \geq 2$, is used); the common colour is 0, so $\chi(x) = 0$ is forced. In regime 2, $x \in [1, t] = [1, d]$, so $\chi(x) = \alpha(x - 1)$ with $x - 1 \in [0, d - 1]$: $\chi(x) = 1 \neq 0$: contradiction. In regime 1, $x \in [1, t] = [1, 2d - r]$, so $x - 1 \in [0, 2d - r - 1]$, all inside $[0, 2d]$; thus $\chi(x) = 0$ requires $x - 1 \in [d, 2d - r - 1]$. If $r = d$ this interval is empty, so $\chi(x) = 1 \neq 0$: contradiction. If $r \leq d - 1$, then $x \in [d + 1, 2d - r]$, and $w(y) - 1 = x + a + d - 1 \equiv (x - 1) + r + d \pmod{2d}$, with $(x - 1) + r + d \in [2d + r, 3d - 1]$; subtracting $2d$, the residues lie in $[r, d - 1]$, a subset of $[0, d - 1]$: $\chi(y) = 1 \neq 0$: contradiction.

All cases being contradictory, χ contains no monochromatic triple. By the parametrisation (1), χ is therefore a 2-colouring of $[1, N^*]$ with no monochromatic solution of $x_1 + ax_2 - x_3 = a - d$; its restriction to $[1, N]$ is such a colouring of $[1, N]$ for every $N \leq N^*$. Combining with monotonicity (Lemma 2.1) gives $\text{Rad}_2(a; a - d) > N^*$, that is,

$$\text{Rad}_2(a; a - d) \geq N^* + 1 = (a + 3)d + a - \min(r - 1, d - 1). \quad \square$$

Remark 1 (sharpness of $d \geq 2$; consistency with the case $d = 1$). For $d = 1$ (where $r = 1$, regime 1, $t = 1$, and the existence condition forces a odd, with $q = (a - 1)/2$) the same construction and the entire argument go through *except* for the single step $\chi(d + 2) = 0$ in Lemma 4.3, used in Case $n = +1$, sub-case $J(y) = d$ of the proof of Theorem 3, which requires $d + 2 \leq 2d$. For $d = 1$ one has instead $\chi(3) = \alpha(2) = 1$ (as $2 \equiv 0 \pmod{2}$), and the triple $(x, k, y) = (1, 3, 2a + 2)$ is monochromatic of colour 1 in the length- $(2a + 2)$ construction: $x = 1 = t$ lies in the initial chunk with $\chi(1) = \alpha(0) = 1$; and $y = 1 + 2a + 1 = 2a + 2 = N^*$ has $J(y) = 1 = d$ and $w(y) = y - a = a + 2$, with $w(y) - 1 = a + 1 \equiv 0 \pmod{2}$ because a is odd, so $\chi(y) = 1$. This is the unique failure mode, and it is consistent with the value $\text{Rad}_2(a; a - 1) = 2a + 2$ established for $d = 1$: one has $2a + 2 = N^*|_{d=1}$, exactly one below the value $((a + 3)d + a - \min(r - 1, d - 1))|_{d=1} = 2a + 3$ that the general formula would predict at $d = 1$. This confirms that the case analysis in the proof of Theorem 3 is tight and that $d \geq 2$ is exactly the needed hypothesis.

Remark 2 (computational verification). All lemmas above and every intermediate inequality in their proofs were verified computationally before being written up. The chunk–offset normal form of Lemma 4.2 ($\chi(x) = \alpha(w(x) - 1)$, with $t = \max(d, 2d - r)$ and jump points $p_j = ja + t$) reproduces the family colouring of Lemma 4.1 exactly, with total length N^* , on all 6830 pairs with $d \geq 2$, $a \leq 120$ and $r \neq 2d$, plus ten larger pairs with a up to 2051 and d up to 499, including $(a, d) = (1001, 7)$, $(997, 30)$, $(1000, 499)$ and $(2051, 50)$; there were zero discrepancies. Every segment-level claim used inside the proof of Lemma 4.2 — the exact segment boundaries S_0, L_i, T_i (regime 1) and $C_0^b, C_0^s, C_i^b, C_i^s, F^b, F^s$ (regime 2), the jump count on each segment, the position of p_i strictly inside the i -th long run, the phase windows, the aligned phase starts 0 and d , and all segment lengths — was checked pointwise on all 4694 valid pairs with $a \leq 100$, with zero errors. The full case analysis of Theorem 3 — the trichotomy $n \in \{-1, 0, 1\}$, the offset windows and forced colours in every case and sub-case, the colours of $1, \dots, d + 2$, the offset ranges, $q \geq 1$ in regime 1, and the absence of any monochromatic triple — was checked by complete enumeration of all in-range triples $(x, k, y = x + a(k - 1) + d)$ for all 2973 pairs comprising every valid pair with $a \leq 80$ together with ten larger pairs including $(101, 50)$, $(127, 63)$, $(211, 105)$, $(333, 5)$ and $(401, 100)$, again with zero errors. Finally, the construction was cross-validated at length exactly N^* against an independent checker that iterates the raw equation $x_1 + ax_2 - x_3 = a - d$ directly (rather than the triple parametrisation), on further

out-of-sample pairs covering both regimes and the edge values $r = 1$ and $r = d + 1$, including (121, 60) (with $N^* = 7560$), (97, 48), (110, 54) and (123, 2): the family colouring was valid at length exactly N^* in every case. The $d = 1$ triples of Remark 1 were likewise confirmed on the actual construction output for $a = 5, 9, 13$ (colours (1, 1, 1)).

5 The upper bound in the regime $r \leq d$

Throughout this section, $d \geq 2$ is fixed and $a \geq 1$ is an integer subject to the existence condition $\nu_2(a) \leq \nu_2(a - d)$, where we adopt the conventions $\nu_2(0) := \infty$ and $\nu_2(-n) := \nu_2(n)$ for $n > 0$ (they are needed because $c = a - d$ may now be zero or negative). We write $a = 2dq + r$ with $q \geq 0$ and $r \in [1, 2d]$ — this decomposition is unique — and we assume the regime condition $r \leq d$. In contrast with the rest of the paper we do *not* assume $a > d$, and the theorem proved here covers *all* $q \geq 0$ at once: $q \geq 1$ holds if and only if $a = 2dq + r \geq 2d + 1$, i.e. if and only if $c = a - d > 0$ (cf. Lemma 2.3(ii)), which is the window $0 < c < a$ of Theorem 2; while $q = 0$ holds if and only if $a = r \leq d$, i.e. if and only if $c \leq 0$ — so the single statement also covers the whole half-line $c \leq 0$, where the regime condition $r \leq d$ is automatic. We prove that every 2-colouring of $[1, N]$, where

$$N := (a + 3)d + a - (r - 1),$$

contains a monochromatic solution of $x_1 + ax_2 - x_3 = a - d$ (Theorem 4). For $q \geq 1$, combined with the lower bound of Section 4, this establishes Theorem 2 in the regime $r \leq d$; for $q = 0$ it is the essential input to the proof of Conjecture 1 of Dwivedi–Tripathi (Theorem 6) in Section 7. All range conditions below are derived from the single inequality $N - E = 2d(q + 1) + 1 \geq 2d + 1$, valid for every $q \geq 0$ (for $q \geq 1$ the stronger $N - E \geq 4d + 1$ is available, but it is never needed), together with the inequality $r \leq a$, which replaces the bound $a \geq 2d + 1$, no longer available at $q = 0$. The proof is a rigidity argument, organised in five moves. A four-triple *transport lemma* (Lemma 5.3) shows that every colour-1 cell propagates the local pattern 1, 0, 1 with step d , as long as there is room on the right. Applied once, transport yields a *pairing* $\chi(s + d) = 1 - \chi(s)$ on $[1, d]$ (Lemma 5.4), which packages the prefix of the colouring into a $2d$ -periodic antisymmetric pattern p . Applied inductively, transport forces χ to agree with p on a long initial *window* (Lemma 5.5), while a complementary argument forces the inverted pattern $\chi(z) = 1 - p(z - E)$ on the *far zone* $[E + 1, N]$, where $E := d(a + 1)$ (Lemma 5.6). The *endgame* splits into two cases according to the restriction $\varphi := \chi|_{[1, d]}$. If $\varphi \not\equiv 0$, the window reaches beyond $a + 2d$, and the solution triples $(a + u, d, E + u)$ with $u \in [1, 2d]$ couple window values to far-zone values along a complete residue system modulo $2d$; a translation-invariance argument in $\mathbb{Z}_{2d}$ — resting on $\gcd(r, 2d) \mid d$, which follows from the existence condition (Lemma 5.1) — then produces a monochromatic triple (Lemma 5.7). If $\varphi \equiv 0$, the single explicit triple $(d + 1 - r, 2, a + 2d + 1 - r)$ is monochromatic (Lemma 5.8).

Setup. Since $q \geq 0$ and $r \geq 1$, we have $a = 2dq + r \geq r$, with $a = r$ if and only if $q = 0$; the inequality $r \leq a$ is the workhorse fact that will stand in for $a \geq 2d + 1$ at $q = 0$. Since $r \leq d$ we have $\min(r - 1, d - 1) = r - 1$, so N is the value asserted by Theorem 2 in this regime, and expanding,

$$N = ad + 3d + a + 1 - r = (d + 1)a + 3d + 1 - r.$$

With $E = d(a + 1)$ this gives

$$N - E = (d + 1)a + 3d + 1 - r - d(a + 1) = a + 2d + 1 - r = 2d(q + 1) + 1 \geq 2d + 1,$$

using $a = 2dq + r$ and $q \geq 0$; in particular $E + 2d \leq N - 1$. For $q = 0$ one has $r = a$, so $N = (d + 1)a + 3d + 1 - a = (a + 3)d + 1 = (a + 3)(a - c) + 1$ with $c = a - d$, which is exactly the value predicted by Conjecture 1 of Dwivedi–Tripathi. By (1), the solutions of $x_1 + ax_2 - x_3 = a - d$ in $[1, N]^3$ are exactly the triples $T(x, k) = (x, k, x + a(k - 1) + d)$ with $x \geq 1$, $k \in [1, N]$ and $x + a(k - 1) + d \leq N$; values may repeat. For $k_0 \in [1, N]$ we write $D_{k_0} := (k_0 - 1)a + d$ for the gap of the triples $T(\cdot, k_0)$; note that $D_{d+1} = da + d = E$. Finally, by Lemma 2.2, in every proof by contradiction below we may and do normalise $\chi(1) = 0$.

We isolate first the 2-adic ingredient announced in Lemma 2.3(iii).

Lemma 5.1 (2-adic lemma). *Let $e := \nu_2(d)$. If $\nu_2(a) \leq \nu_2(a - d)$ — with the conventions $\nu_2(0) = \infty$ and $\nu_2(-n) = \nu_2(n)$, since $a - d$ may be zero or negative when $q = 0$ — then $\nu_2(r) \leq e$; consequently $\gamma := \gcd(r, 2d)$ divides d .*

Proof. Write $d = 2^e m$ with m odd, and suppose for contradiction that $\nu_2(r) \geq e + 1$. Then $r \neq d$, since $\nu_2(d) = e < e + 1$; write $r = 2^{e+1} r'$. First, $\nu_2(a) \geq e + 1$: in $a = 2dq + r$ the term $2dq$ either vanishes (when $q = 0$, with $\nu_2(0) = \infty \geq e + 1$) or satisfies $\nu_2(2dq) = e + 1 + \nu_2(q) \geq e + 1$, and $\nu_2(r) \geq e + 1$, so 2^{e+1} divides both terms. (For $q = 0$ this is simply $\nu_2(a) = \nu_2(r) \geq e + 1$.) Second, $r - d = 2^{e+1} r' - 2^e m = 2^e(2r' - m)$ with $2r' - m$ odd and nonzero, so $\nu_2(r - d) = e$ exactly — whether $r - d$ is positive or negative, by the convention $\nu_2(-n) = \nu_2(n)$. Hence $a - d = 2dq + (r - d)$ is the sum of a term of 2-adic valuation at least $e + 1$ (or of the zero term, when $q = 0$) and a term of valuation exactly e , so by the ultrametric inequality with strict disparity $\nu_2(a - d) = e$ exactly; in particular $a - d \neq 0$, so $\nu_2(a - d)$ is an ordinary integer. (For $q = 0$ this is simply $a - d = r - d$, of valuation e .) Then $\nu_2(a) \geq e + 1 > e = \nu_2(a - d)$, contradicting the hypothesis. Therefore $\nu_2(r) \leq e$.

In the boundary case $r = d$ and $q = 0$ — that is, $a = d$ and $a - d = 0$ — the hypothesis holds trivially ($\nu_2(0) = \infty$) and so does the conclusion ($\nu_2(r) = \nu_2(d) = e \leq e$); the contradiction argument above never encounters $r = d$.

For the consequence, consider $\gamma = \gcd(r, 2d)$. Its 2-part satisfies $\nu_2(\gamma) = \min(\nu_2(r), \nu_2(2d)) = \nu_2(r) \leq e$, since $\nu_2(r) \leq e < e + 1 = \nu_2(2d)$. Its odd part divides the odd part of $2d$, which is m . A positive integer divides $d = 2^e m$ if and only if its 2-adic valuation is at most e and its odd part divides m ; γ satisfies both conditions, so $\gamma \mid d$. $\square$

Lemma 5.2 (first forced colour). *Let χ be a valid colouring of $[1, N]$ with $\chi(1) = 0$. Then $\chi(d + 1) = 1$.*

Proof. The triple $T(1, 1) = (1, 1, 1 + d)$ is a solution triple, and $1 + d \leq N$ since $N = E + 2d(q + 1) + 1 \geq E + 2d + 1 = d(a + 1) + 2d + 1 \geq 3d + 1 > d + 1$, using only $a \geq 1$ and $q \geq 0$. Its first two entries have colour $\chi(1) = 0$; if $\chi(d + 1) = 0$ the triple would be monochromatic. Hence $\chi(d + 1) = 1$. $\square$

Lemma 5.3 (transport). *Let χ be a valid colouring of $[1, N]$ with $\chi(1) = 0$. Let $k_0 \in [1, N]$ satisfy $\chi(k_0) = 1$ and put $D := D_{k_0} = (k_0 - 1)a + d$. Then for every $y \in [1, N - D - d]$ with $\chi(y) = 1$,*

$$\chi(y + d) = 0 \quad \text{and} \quad \chi(y + 2d) = 1.$$

Proof. This lemma and its proof are independent of q . All four triples used below are solution triples inside $[1, N]^3$; the range checks are given in brackets.

Step 1. $T(y, k_0) = (y, k_0, y + D)$ [$x_3 = y + D \leq N - d \leq N$]. Since $\chi(y) = \chi(k_0) = 1$, monochromaticity would follow from $\chi(y + D) = 1$; hence $\chi(y + D) = 0$.

Step 2. $T(y + D, 1) = (y + D, 1, y + D + d)$ [$x_3 = y + D + d \leq N$ by the hypothesis $y \leq N - D - d$]. Since $\chi(y + D) = \chi(1) = 0$, we get $\chi(y + D + d) = 1$.

Step 3. $T(y + d, k_0) = (y + d, k_0, y + d + D)$ [$x_1 = y + d \geq 1$; $x_3 = y + D + d \leq N$]. Since $\chi(y + d + D) = 1 = \chi(k_0)$, monochromaticity would follow from $\chi(y + d) = 1$; hence $\chi(y + d) = 0$.

Step 4. $T(y + d, 1) = (y + d, 1, y + 2d)$ [$x_3 = y + 2d \leq y + D + d \leq N$, using $D = (k_0 - 1)a + d \geq d$]. Since $\chi(y + d) = \chi(1) = 0$, we get $\chi(y + 2d) = 1$.

(Recall that repeated values among x_1, x_2, x_3 are permitted, so no genericity conditions are needed.) $\square$

Lemma 5.4 (pairing). *Let χ be valid on $[1, N]$ with $\chi(1) = 0$. Then $\chi(s + d) = 1 - \chi(s)$ for every $s \in [1, d]$. Define $\varphi(s) := \chi(s)$ for $s \in [1, d]$ (so $\varphi(1) = 0$), and define $p: \mathbb{Z} \rightarrow \{0, 1\}$ by*

$$p(u) = \begin{cases} \varphi(s), & u \equiv s \pmod{2d} \text{ with } s \in [1, d], \\ 1 - \varphi(s), & u \equiv s + d \pmod{2d} \text{ with } s \in [1, d]. \end{cases}$$

Then p is $2d$ -periodic and antisymmetric, i.e. $p(u + d) = 1 - p(u)$ for all u , with $p(1) = 0$, and $\chi(u) = p(u)$ for all $u \in [1, 2d]$.

Proof. Fix $s \in [1, d]$. *Not both 0:* the triple $T(s, 1) = (s, 1, s + d)$ lies in $[1, N]^3$, since $s + d \leq 2d < N$ (indeed $N \geq E + 2d + 1 > 2d$); if $\chi(s) = \chi(s + d) = 0$, then together with $\chi(1) = 0$ it would be monochromatic — contradiction. *Not both 1:* apply the transport lemma (Lemma 5.3) with $k_0 = d + 1$ (legitimate since $\chi(d + 1) = 1$ by Lemma 5.2, and $d + 1 \leq N$), $D = D_{d+1} = E$, at $y = s$. The room condition $s \leq N - E - d$ holds for every $q \geq 0$:

$$N - E - d = 2d(q + 1) + 1 - d = 2dq + d + 1 \geq d + 1 > d \geq s.$$

(At $q = 0$ the room is exactly $d + 1$, exceeding the largest seed $s = d$ by exactly one unit; this is the tightest point of the pairing step, and it is one of the two places where a single unit of room is lost at length $N - 1$.) If $\chi(s) = 1$, transport yields $\chi(s + d) = 0$. Hence exactly one of $\chi(s), \chi(s + d)$ equals 1, i.e. $\chi(s + d) = 1 - \chi(s)$.

The stated properties of p follow directly. It is $2d$ -periodic by construction. Antisymmetry: if $u \equiv s \pmod{2d}$ with $s \in [1, d]$, then $u + d \equiv s + d$ and $p(u + d) = 1 - \varphi(s) = 1 - p(u)$; if $u \equiv s + d$, then $u + d \equiv s \pmod{2d}$ and $p(u + d) = \varphi(s) = 1 - (1 - \varphi(s)) = 1 - p(u)$. Finally, $\chi(u) = p(u)$ on $[1, 2d]$ is exactly the definition of φ together with the pairing just proved. $\square$

Lemma 5.5 (window). *Let χ be valid on $[1, N]$ with $\chi(1) = 0$, and let φ, p be as in Lemma 5.4. Define*

$$s^* := \min\{s \in [2, d + 1] : p(s) = 1\}, \quad D^* := (s^* - 1)a + d, \quad X^* := N - D^*,$$

where s^ is well defined since $p(d + 1) = 1 - \varphi(1) = 1$. Then*

$$\chi(u) = p(u) \quad \text{for every } u \in [1, X^*].$$

Moreover $X^ \geq N - E = 2d(q + 1) + 1 \geq 2d + 1$, since $D^* \leq D_{d+1} = E$.*

Proof. First, $\chi(s^*) = 1$: if $s^* \leq d$ this is $\varphi(s^*) = p(s^*) = 1$ by the definition of φ ; if $s^* = d + 1$ it is Lemma 5.2. Also $s^* \leq d + 1 \leq N$. The transport lemma therefore applies with $k_0 = s^*$, gap $D = D^*$, and room condition $y \leq N - D^* - d = X^* - d$.

Fix $s \in [1, d]$, and let $y_0 := s$ if $\varphi(s) = 1$, and $y_0 := s + d$ otherwise. By Lemma 5.4, $\chi(y_0) = 1$ and $y_0 \in [1, 2d]$.

Claim (induction on $j \geq 0$):

- (a) if $y_0 + 2dj \leq X^* + d$, then $\chi(y_0 + 2dj) = 1$;
- (b) if $y_0 + 2dj + d \leq X^*$, then $\chi(y_0 + 2dj + d) = 0$.

Base case $j = 0$. Part (a) is the seed $\chi(y_0) = 1$ (note $y_0 \leq 2d \leq X^* + d$, since $X^* \geq 2d + 1 > d$). For (b): if $y_0 + d \leq X^*$, then $y_0 \leq X^* - d$, so transport at $y = y_0$ gives $\chi(y_0 + d) = 0$. *Inductive step.* Assume (a) at j , i.e. $\chi(y_j) = 1$ for $y_j := y_0 + 2dj \leq X^* + d$ (if the hypothesis of (a) at j fails, then the hypotheses of (a) at all larger indices and of (b) at all indices $\geq j$ fail as well, and there is nothing to prove). For (b) at j : if $y_j + d \leq X^*$, then $y_j \leq X^* - d$, and transport at y_j gives $\chi(y_j + d) = 0$. For (a) at $j + 1$: if $y_j + 2d \leq X^* + d$, then $y_j \leq X^* - d$, and transport at y_j gives $\chi(y_j + 2d) = 1$. This proves the claim.

Coverage. Let $u \in [1, X^*]$. Its residue modulo $2d$ equals that of y_0 or of $y_0 + d$ for the appropriate $s \in [1, d]$, since $\{y_0, y_0 + d\} \equiv \{s, s + d\} \pmod{2d}$.

Case $u \equiv y_0 \pmod{2d}$. Since $y_0 \in [1, 2d]$, the least positive representative of its class is y_0 itself, so $u = y_0 + 2dj$ for some $j \geq 0$, with $u \leq X^* < X^* + d$; by (a), $\chi(u) = 1 = p(u)$, since $p(u) = p(y_0) = \chi(y_0) = 1$ by construction.

Case $u \equiv y_0 + d \pmod{2d}$. If $y_0 = s$ (the case $\varphi(s) = 1$), the least positive representative is $s + d = y_0 + d$, so $u = y_0 + d + 2dj$ for some $j \geq 0$ with $u \leq X^*$; by (b), $\chi(u) = 0 = p(u)$, since $p(u) = p(s + d) = 1 - \varphi(s) = 0$. If $y_0 = s + d$ (the case $\varphi(s) = 0$), the least positive representative is s : either $u = s$, a seed value with $\chi(s) = \varphi(s) = 0 = p(u)$ (Lemma 5.4), or $u = s + 2dj = y_0 + d + 2d(j - 1)$ for some $j \geq 1$ with $u \leq X^*$, and (b) gives $\chi(u) = 0 = p(u)$.

In all cases $\chi(u) = p(u)$. Every step above is independent of q ; the only q -sensitive quantity is the guaranteed window length $X^* \geq 2d(q + 1) + 1$, which at $q = 0$ equals $2d + 1$ and in particular still covers $[1, 2d]$ and the cell $2d + 1$. $\square$

Lemma 5.6 (far zone). *Let χ be valid on $[1, N]$ with $\chi(1) = 0$, let p, s^*, X^* be as above, and $E = d(a + 1)$. Then*

$$\chi(z) = 1 - p(z - E) \quad \text{for every } z \in [E + 1, N].$$

Proof. Let $u := z - E \in [1, N - E]$, where $N - E = 2d(q + 1) + 1$. Since $u \leq N - E \leq X^*$ (Lemma 5.5), the value $\chi(u) = p(u)$ is known.

Case 1: $p(u) = 1$. Then $\chi(u) = 1$, and the triple $T(u, d + 1) = (u, d + 1, u + E) = (u, d + 1, z)$ lies in $[1, N]^3$ ($x_2 = d + 1 \leq N$; $x_3 = z \leq N$ by hypothesis). Since $\chi(u) = \chi(d + 1) = 1$ (Lemma 5.2), monochromaticity would follow from $\chi(z) = 1$; hence $\chi(z) = 0 = 1 - p(u)$.

Case 2: $p(u) = 0$. Sub-case $u \leq N - E - d$: then $u + d \leq N - E$ and $p(u + d) = 1$ by antisymmetry, so Case 1 applied to $u + d$ gives $\chi(z + d) = \chi(E + u + d) = 0$. The triple $T(z, 1) = (z, 1, z + d)$ lies in $[1, N]^3$ ($z + d = E + u + d \leq N$). If $\chi(z) = 0$, then $\chi(z) = \chi(1) = \chi(z + d) = 0$ would be monochromatic — contradiction; hence $\chi(z) = 1 = 1 - p(u)$.

Sub-case $u > N - E - d$, i.e. $u \geq N - E - d + 1 = 2dq + d + 2$: since $2dq + d + 2 \geq d + 2 > d + 1$ for every $q \geq 0$, we have $u - d \geq 2 \geq 1$, and $p(u - d) = 1 - p(u) = 1$ by antisymmetry. Moreover $u - d \leq N - E$, so Case 1 applied to $u - d$ gives $\chi(z - d) = \chi(E + (u - d)) = 0$. Also $z - d = E + (u - d) \geq E + 1$, so $z - d \in [E + 1, N]$ and the triple $T(z - d, 1) = (z - d, 1, z)$ lies in $[1, N]^3$ ($x_3 = z \leq N$). If $\chi(z) = 0$, then $\chi(z - d) = \chi(1) = \chi(z) = 0$ would be monochromatic — contradiction; hence $\chi(z) = 1 = 1 - p(u)$.

(For $q = 0$ the sub-case threshold reads $u \geq d + 2$, with u ranging over $[1, 2d + 1]$: both sub-cases are nonempty and together they exhaust the range; at the top cell $u = 2d + 1 \equiv 1 \pmod{2d}$ one has $p(u) = p(1) = 0$ and $u - d = d + 1$ with $p(d + 1) = 1$, as used.) $\square$

Remark 3 (window–far-zone overlap). When $X^* \geq E + 1$ the window of Lemma 5.5 and the far zone of Lemma 5.6 overlap, and on the overlap the two forced formulas may disagree. This is no inconsistency: both lemmas are implications from the assumed validity of χ , so any disagreement refutes validity outright; the endgame below provides a uniform contradiction in all cases regardless.

Lemma 5.7 (endgame, Case A: $\varphi \neq 0$). *Suppose χ is a valid colouring of $[1, N]$ with $\chi(1) = 0$, and suppose $s^* \leq d$ (that is, $\varphi(s) = 1$ for some $s \in [2, d]$; this requires $d \geq 2$). Set $\varepsilon := \varphi(d)$. Then there exists $u \in [1, 2d]$ with*

$$p(u + r) = \varepsilon \quad \text{and} \quad p(u) = 1 - \varepsilon,$$

and for any such u the solution triple $(a + u, d, E + u)$ is monochromatic of colour ε . This contradicts validity, so no such χ exists.

Proof. Existence of u . (This step is independent of q and does not use the hypothesis $s^* \leq d$, which enters only through the forced colours below.) View p as a function on $\mathbb{Z}_{2d}$ (residues represented by $[1, 2d]$) and let $P := \{v \in \mathbb{Z}_{2d} : p(v) = \varepsilon\}$. Suppose no u as stated exists; then for every $u \in \mathbb{Z}_{2d}$, $p(u + r) = \varepsilon$ implies $p(u) = \varepsilon$, i.e. $P - r \subseteq P$, where $P - r := \{v : v + r \in P\}$. Translation by r is a bijection of $\mathbb{Z}_{2d}$, so $|P - r| = |P|$, forcing $P - r = P$. Iterating, $P - jr = P$ for all $j \geq 0$, so P is invariant under translation by every element of the subgroup $\langle r \rangle = \gamma(\mathbb{Z}_{2d})$, where $\gamma = \gcd(r, 2d)$. By Lemma 5.1, $\gamma \mid d$, so $d \in \langle r \rangle$ and $P - d = P$. Now $d \in P$, because $p(d) = \varphi(d) = \varepsilon$; hence $2d = d + d \in P$ (using $P = P + d$), i.e. $p(2d) = \varepsilon$. But antisymmetry gives $p(2d) = p(d + d) = 1 - p(d) = 1 - \varepsilon$ — a contradiction. The required u exists.

The triple. Fix such a $u \in [1, 2d]$ and consider

$$T(a + u, d) = (a + u, d, (a + u) + (d - 1)a + d) = (a + u, d, E + u).$$

Ranges: $x_1 = a + u \geq a + 1 \geq 2$; $x_2 = d \in [1, N]$; $x_3 = E + u \leq E + 2d \leq N$, since $N - E = 2d(q + 1) + 1 \geq 2d + 1 > 2d$ for every $q \geq 0$. Forced colours:

(1) $x_1 = a + u \leq a + 2d \leq X^*$: indeed $s^* \leq d$ gives $D^* = (s^* - 1)a + d \leq (d - 1)a + d$, so

$$X^* = N - D^* \geq N - (d - 1)a - d = (d + 1)a + 3d + 1 - r - (d - 1)a - d = 2a + 2d + 1 - r,$$

and $(2a + 2d + 1 - r) - (a + 2d) = a + 1 - r \geq 1 > 0$, because $r \leq a$ (Setup; valid for every $q \geq 0$, with equality $r = a$ exactly at $q = 0$, where the slack is exactly one unit). Hence, by the window lemma, $\chi(a + u) = p(a + u) = p(u + r)$, because $a - r = 2dq \equiv 0 \pmod{2d}$ — an identity valid at $q = 0$ as well, where $a = r$ exactly; by the choice of u , $\chi(a + u) = \varepsilon$.

(2) $\chi(d) = \varphi(d) = \varepsilon$ by definition.

(3) $x_3 = E + u \in [E + 1, E + 2d] \subseteq [E + 1, N]$, so by the far-zone lemma $\chi(E + u) = 1 - p(u) = 1 - (1 - \varepsilon) = \varepsilon$.

Thus all three entries have colour ε : the triple is monochromatic, contradicting the validity of χ . $\square$

Lemma 5.8 (endgame, Case B: $\varphi \equiv 0$). *Suppose χ is a valid colouring of $[1, N]$ with $\chi(1) = 0$, and suppose $s^* = d + 1$ (that is, $\varphi(s) = 0$ for all $s \in [1, d]$). Then the solution triple $(d + 1 - r, 2, a + 2d + 1 - r)$ is monochromatic of colour 0. This contradicts validity, so no such χ exists.*

Proof. Here $D^* = D_{d+1} = E$, so $X^* = N - E = a + 2d + 1 - r = 2d(q + 1) + 1$. Consider

$$T(d + 1 - r, 2) = (d + 1 - r, 2, (d + 1 - r) + a + d) = (d + 1 - r, 2, a + 2d + 1 - r).$$

Ranges: $x_1 = d + 1 - r \geq 1$ since $r \leq d$ (at $q = 0$ this reads $x_1 = d + 1 - a \geq 1$, using $r = a \leq d$); $x_2 = 2 \in [1, N]$; $x_3 = a + 2d + 1 - r = X^* \leq N$ since $D^* \geq 0$. Forced colours:

- (1) $\chi(d + 1 - r) = \varphi(d + 1 - r) = 0$, since $d + 1 - r \in [1, d]$ (as $1 \leq r \leq d$) and $\varphi \equiv 0$;
- (2) $\chi(2) = \varphi(2) = 0$, using $d \geq 2$ so that $2 \in [1, d]$;
- (3) $x_3 = X^*$ lies in the window, and $X^* - 1 = a + 2d - r = 2dq + 2d = 2d(q + 1) \equiv 0 \pmod{2d}$ — an identity valid verbatim at $q = 0$, where $X^* - 1 = 2d$ — so $X^* \equiv 1 \pmod{2d}$ and, by the window lemma and the $2d$ -periodicity of p , $\chi(X^*) = p(X^*) = p(1) = \varphi(1) = 0$.

All three entries have colour 0: the triple is monochromatic, contradicting the validity of χ . Every inequality in this proof is independent of q . $\square$

Theorem 4 (unified upper bound, regime $r \leq d$, all $q \geq 0$). *Let $d \geq 2$ and let $a \geq 1$ satisfy $\nu_2(a) \leq \nu_2(a - d)$ (with the conventions $\nu_2(0) = \infty$ and $\nu_2(-n) = \nu_2(n)$); write $a = 2dq + r$ with $q \geq 0$ and $r \in [1, 2d]$, and assume $r \leq d$. Then every 2-colouring of $[1, N]$, where $N = (a + 3)d + a - (r - 1) = (d + 1)a + 3d + 1 - r$, contains a monochromatic solution of $x_1 + ax_2 - x_3 = a - d$. Consequently*

$$\text{Rad}_2(a; a - d) \leq (a + 3)d + a - (r - 1) = (a + 3)d + a - \min(r - 1, d - 1).$$

Proof. Suppose some 2-colouring χ of $[1, N]$ has no monochromatic solution. By Lemma 2.2 we may assume $\chi(1) = 0$. Lemma 5.2 gives $\chi(d + 1) = 1$; Lemma 5.4 defines φ on $[1, d]$ and its $2d$ -periodic antisymmetric extension p , with $\chi = p$ on $[1, 2d]$; and $s^* = \min\{s \in [2, d + 1] : p(s) = 1\}$ is well defined since $p(d + 1) = 1 - \varphi(1) = 1$ (Lemma 5.5). If $s^* \leq d$, Lemma 5.7 produces the monochromatic solution triple $(a + u, d, E + u)$ — contradiction. If $s^* = d + 1$, then $p(s) = 0$ for all $s \in [2, d]$, and $\varphi(1) = 0$, so $\varphi \equiv 0$ on $[1, d]$; Lemma 5.8 produces the monochromatic triple $(d + 1 - r, 2, a + 2d + 1 - r)$ — contradiction. Hence no such χ exists and $\text{Rad}_2(a; a - d) \leq N$. Since $r \leq d$ gives $\min(r - 1, d - 1) = r - 1$, this is exactly the stated bound. The chain of lemmas used only $N - E = 2d(q + 1) + 1 \geq 2d + 1$ (Lemmas 5.2, 5.4, 5.6 and 5.7), $r \leq a$ (Lemma 5.7), $r \leq d$ (Lemma 5.8), $d \geq 2$ (Lemma 5.8, and the nonvacuity of Case A), and $\gcd(r, 2d) \mid d$ from the existence condition (Lemmas 5.1 and 5.7) — each established for every $q \geq 0$. $\square$

Remark 4 (tightness). At length $N - 1$ the same derivations lose exactly one unit of room: the window lemma then forces $\chi = p$ only on $[1, X^* - 1]$, and at $q = 0$ the pairing room condition $s \leq N - E - d$ and the Case A window slack $a + 1 - r$ are tight with margin exactly one. For $q \geq 1$ the extremal family of Section 4, which realises $\varphi \equiv 0$ and places its d long runs beyond the shortened window, shows that the bound of Theorem 4 cannot be improved; for $q = 0$ the valid colourings of $[1, N - 1]$ supplied by the lower bound of Dwivedi–Tripathi [1] (Theorem 2 there, valid for all $c < a$), invoked in Section 7, show the same (cf. Lemma 2.1).

6 The upper bound in the regime $r > d$

The proof is a chain of forcing steps. Two elementary *pair constraints*, coming from the multipliers $k = 1$ and $k = d + 1$ (Lemma 6.1), compose into a $\pm 2d$ ladder showing that any putative valid colouring is $2d$ -periodic on an initial *window* (Lemma 6.2) and colour-*flipped* on a *top block* of the same length (Lemma 6.3); the window pattern $\phi: \mathbb{Z}_{2d} \rightarrow \{0, 1\}$ is then *antipodal*, $\phi(\rho + d) = 1 - \phi(\rho)$ (Lemma 6.4). Next, three explicit families of conflict triples — with multipliers $k = d + 2$, $k = d$, and a two-triple dichotomy using the consecutive multipliers $k = i, i + 1$ through a common cell — forbid every *s-plateau* of ϕ (Lemma 6.5), and a purely combinatorial case analysis upgrades this to *s-antiperiodicity*, $\phi(m + s) = 1 - \phi(m)$ (Lemma 6.6). Finally, a 2-adic computation shows that antipodality and *s-antiperiodicity* are incompatible exactly when $\nu_2(s) \neq \nu_2(d)$ — and this is precisely the existence condition in this regime (Lemma 6.7) — so no valid colouring exists. The final obstruction is thus *exactly* the 2-adic existence condition: the *s-antiperiodic* patterns are the unique family of colourings escaping the plateau conflicts, and the existence condition is precisely what rules them out.

Setup and standing assumption. Fix $d \geq 2$ and $a > d$ in the regime $r > d$: by Lemma 2.3, under the existence condition $\nu_2(a) \leq \nu_2(a - d)$ we have $r \in [d + 1, 2d - 1]$, so we write

$$a = 2dq + d + s, \quad s := r - d \in [1, d - 1], \quad q \geq 0,$$

and set $c = a - d$. Put

$$N := (a + 3)d + a - (d - 1) = (d + 1)a + 2d + 1, \quad E := d(a + 1), \quad W := a + d + 1.$$

Then $N = E + W$: indeed $(a + 3)d + a - (d - 1) = ad + 3d + a - d + 1 = (d + 1)a + 2d + 1$ and $E + W = d(a + 1) + a + d + 1 = (d + 1)a + 2d + 1 = N$. Moreover $a \equiv d + s \pmod{2d}$, and $a \geq d + 1$ (as $q \geq 0$ and $s \geq 1$), so $W = a + d + 1 \geq 2d + 2 > 2d$. (For $q \geq 1$ one has $a \geq 3d + 1$; the proofs below use only $a \geq d + 1$.) By (1), the solutions of $x_1 + ax_2 - x_3 = c$ in $[1, N]^3$ are exactly the triples $T(x, k) = (x, k, x + a(k - 1) + d)$ with $x, k \geq 1$ and $x + a(k - 1) + d \leq N$ (entries may repeat).

Suppose, for contradiction, that $\chi: [1, N] \rightarrow \{0, 1\}$ is a valid colouring, i.e. admits no monochromatic solution; by Lemma 2.2 we may and do assume $\chi(1) = 0$. This *standing assumption* — a valid colouring χ of $[1, N]$ with $\chi(1) = 0$, together with the quantities N, E, W above — remains in force throughout Lemmas 6.1–6.5; Lemmas 6.6 and 6.7 are self-contained statements about patterns on $\mathbb{Z}_{2d}$ and 2-adic valuations.

Lemma 6.1 (pair constraints). (a) $\chi(d + 1) = 1$.

(b) (R0) There is no $x \in [1, N - d]$ with $\chi(x) = \chi(x + d) = 0$.

(c) (R1) There is no $x \in [1, W]$ with $\chi(x) = \chi(x + E) = 1$.

Proof. (b) For $x \in [1, N - d]$, $T(x, 1) = (x, 1, x + d)$ is a solution ($x_3 = x + a \cdot 0 + d$). If $\chi(x) = \chi(x + d) = 0$, then, since $\chi(1) = 0$, the triple $T(x, 1)$ is monochromatic of colour 0 — contradiction.

(a) Apply (b) with $x = 1$: $T(1, 1) = (1, 1, 1 + d)$ is a solution ($1 + d \leq N$), and $\chi(1) = 0$, so $\chi(d + 1) = 0$ would make it monochromatic; hence $\chi(d + 1) = 1$.

(c) For $x \in [1, W]$, $T(x, d + 1) = (x, d + 1, x + E)$ is a solution: $x + a((d + 1) - 1) + d = x + ad + d = x + d(a + 1) = x + E$, and $x + E \leq W + E = N$. If $\chi(x) = \chi(x + E) = 1$, then, since $\chi(d + 1) = 1$ by (a), the triple $T(x, d + 1)$ is monochromatic of colour 1 — contradiction. $\square$

We refer to the constraints of Lemma 6.1(b) and (c) as R0 and R1.

Lemma 6.2 (window periodicity). *There is a function $\phi: \mathbb{Z}_{2d} \rightarrow \{0, 1\}$ such that $\chi(x) = \phi(x \bmod 2d)$ for every $x \in [1, W]$. Moreover $\phi(1) = 0$ and $\phi(d+1) = 1$.*

Proof. *Claim U (up-step):* if $\chi(x) = 0$ and $1 \leq x \leq W - 2d$, then $\chi(x + 2d) = 0$. Indeed: (i) $\chi(x) = 0$ and $x + d \leq W \leq N$ imply $\chi(x + d) = 1$ by R0. (ii) $x + d \in [1, W]$, so R1 gives $\chi(x + d + E) = 0$. (iii) Apply R0 at position $x + d + E$: it lies in $[1, N - d]$ because $(x + d + E) + d = x + 2d + E \leq W + E = N$ (using $x \leq W - 2d$); since $\chi(x + d + E) = 0$, R0 gives $\chi(x + 2d + E) = 1$. (iv) Apply R1 at position $x + 2d \in [1, W]$: since $\chi(x + 2d + E) = 1$, R1 forces $\chi(x + 2d) = 0$. This proves Claim U.

Claim D (down-step): if $\chi(y) = 0$ and $2d + 1 \leq y \leq W$, then $\chi(y - 2d) = 0$. Indeed: (i) R0 at $y - d$ (note $y - d \geq d + 1 \geq 1$ and $(y - d) + d = y \leq W \leq N$): $\chi(y) = 0$ forces $\chi(y - d) = 1$. (ii) R1 at $y - d \in [1, W]$: $\chi(y - d + E) = 0$. (iii) R0 at $y - 2d + E$ (note $y - 2d + E \geq E + 1 \geq 1$ and $(y - 2d + E) + d = y - d + E \leq W + E = N$): $\chi(y - d + E) = 0$ forces $\chi(y - 2d + E) = 1$. (iv) R1 at $y - 2d \in [1, W]$: $\chi(y - 2d + E) = 1$ forces $\chi(y - 2d) = 0$. This proves Claim D.

Class constancy. Fix $\rho \in [1, 2d]$. The members of the residue class $\{x \equiv \rho \pmod{2d}\}$ inside $[1, W]$ are $z_j = \rho + 2dj$, $j = 0, \dots, \ell$, where $\ell := \lfloor (W - \rho)/(2d) \rfloor$; the list is nonempty since $\rho \leq 2d < W$ (recall $W \geq 2d + 2$). Suppose some member z_i has $\chi(z_i) = 0$. Applying Claim D repeatedly from z_i (each intermediate member is $\leq W$, and $\geq 2d + 1$ while $j \geq 1$) gives $\chi(z_j) = 0$ for all $j \leq i$; in particular $\chi(z_0) = \chi(\rho) = 0$. Applying Claim U repeatedly from z_0 (the step from z_j to z_{j+1} needs $z_j \leq W - 2d$, which holds exactly when $z_{j+1} \leq W$, i.e. whenever z_{j+1} is a member) gives $\chi(z_j) = 0$ for all $j \leq \ell$. Hence either all members are 0 or none is: χ is constant on each residue class inside $[1, W]$. Define $\phi(\rho)$ to be that constant. Then $\phi(1) = \chi(1) = 0$, and $\phi(d+1) = \chi(d+1) = 1$ by Lemma 6.1(a) (both positions lie in $[1, W]$). $\square$

Lemma 6.3 (flip on the top block). *For every $x \in [1, W]$: $\chi(E + x) = 1 - \chi(x)$. Consequently $\chi(E + x) = 1 - \phi(x \bmod 2d)$ on the top block $[E + 1, E + W] = [E + 1, N]$, which together with Lemma 6.2 determines χ on $[1, W] \cup [E + 1, N]$.*

Proof. *Case $\chi(x) = 1$:* R1 at $x \in [1, W]$ directly forbids $\chi(x + E) = 1$, so $\chi(E + x) = 0$.

Case $\chi(x) = 0$ and $x \leq a + 1$: (i) $x + d \leq a + d + 1 = W \leq N$, so R0 gives $\chi(x + d) = 1$. (ii) R1 at $x + d \in [1, W]$ gives $\chi(x + d + E) = 0$. (iii) Apply R0 at position $x + E$: it lies in $[1, N - d]$ because $(x + E) + d = x + d + E \leq (a + 1) + d + E = N$; since $\chi(x + d + E) = 0$, R0 forbids $\chi(x + E) = 0$; hence $\chi(x + E) = 1$.

Case $\chi(x) = 0$ and $a + 2 \leq x \leq W$: (i) $x - d \geq a + 2 - d \geq 1$ (as $a \geq d + 1$), and R0 at $x - d$ forbids $\chi(x - d) = 0 = \chi(x)$; hence $\chi(x - d) = 1$. (ii) R1 at $x - d \in [1, W]$ gives $\chi(x - d + E) = 0$. (iii) R0 at $x - d + E$ (note $(x - d + E) + d = x + E \leq W + E = N$) forbids $\chi(x + E) = 0$; hence $\chi(x + E) = 1$.

In all cases $\chi(E + x) = 1 - \chi(x)$; by Lemma 6.2, $\chi(x) = \phi(x \bmod 2d)$ for $x \in [1, W]$. $\square$

Lemma 6.4 (antipodality of the window pattern). $\phi(\rho + d) = 1 - \phi(\rho)$ for every $\rho \in \mathbb{Z}_{2d}$.

Proof. It suffices to prove the identity for $\rho \in [1, d]$, since the pairs $\{\rho, \rho + d\}$ partition $\mathbb{Z}_{2d}$. For such ρ , both positions ρ and $\rho + d$ lie in $[1, 2d]$, and $2d \leq W$, so $\chi(\rho) = \phi(\rho)$ and $\chi(\rho + d) = \phi(\rho + d)$.

Not both 0: if $\phi(\rho) = \phi(\rho + d) = 0$, then $\chi(\rho) = \chi(\rho + d) = 0$ with $\rho \leq N - d$, contradicting R0.

Not both 1: if $\phi(\rho) = \phi(\rho+d) = 1$, then by Lemma 6.3, $\chi(E+\rho) = 0$ and $\chi(E+\rho+d) = 0$ (both ρ and $\rho+d$ lie in $[1, W]$). The position $E+\rho$ lies in $[1, N-d]$, since $E+\rho+d \leq E+2d \leq E+W = N$. So R0 is violated at $E+\rho$ — contradiction.

Hence exactly one of $\phi(\rho)$, $\phi(\rho+d)$ is 0, i.e. $\phi(\rho+d) = 1 - \phi(\rho)$. $\square$

Lemma 6.5 (plateau-conflict triples). *Under the standing assumption (a valid colouring χ of $[1, N]$ with $\chi(1) = 0$) and with the objects of Lemmas 6.2–6.4, call $m \in [1, d+1]$ an s -plateau of colour κ if $\phi(m) = \phi(m+s) = \kappa$ (indices modulo $2d$; note $m+s \leq 2d$, so no wrap occurs). Then:*

(V) *there is no s -plateau of colour $1 - \phi(2)$;*

(IV) *there is no s -plateau of colour $1 - \phi(d)$;*

(Cb) *there is no pair (i, m) with $i \in [1, d-1]$, $\phi(i) \neq \phi(i+1)$, and m an s -plateau of colour $\phi(i+1)$.*

Proof. Throughout the proof we use: $\chi(x) = \phi(x \bmod 2d)$ on $[1, W]$ (Lemma 6.2); $\chi(E+x) = 1 - \phi(x \bmod 2d)$ for $x \in [1, W]$ (Lemma 6.3); $\phi(\rho+d) = 1 - \phi(\rho)$ (Lemma 6.4); and $a \equiv d+s \pmod{2d}$, whence $\phi(a+m) = \phi(d+s+m) = 1 - \phi(m+s)$.

(V): Suppose $\phi(m) = \phi(m+s) = 1 - \phi(2) =: \kappa$ for some $m \in [1, d+1]$. Consider the triple $T(m, d+2) = (m, d+2, E+a+m)$. It is a solution: $m+a((d+2)-1)+d = m+(d+1)a+d = m+E+a$, and $E+a+m \leq E+a+d+1 = N$; all entries are ≥ 1 . Colours: $\chi(m) = \phi(m) = \kappa$ (as $m \leq d+1 \leq W$); $\chi(d+2) = \phi(d+2) = 1 - \phi(2) = \kappa$ (as $d+2 \leq W$); $\chi(E+(a+m)) = 1 - \phi(a+m) = 1 - (1 - \phi(m+s)) = \phi(m+s) = \kappa$ (as $a+m \leq a+d+1 = W$, so Lemma 6.3 applies). The triple is monochromatic — contradiction.

(IV): Suppose $\phi(m) = \phi(m+s) = 1 - \phi(d)$ for some $m \in [1, d+1]$. Consider the triple $T(a+m, d) = (a+m, d, E+m)$. It is a solution: $(a+m)+a(d-1)+d = da+d+m = E+m \leq N$; all entries are ≥ 1 . Colours: $\chi(a+m) = \phi(a+m) = 1 - \phi(m+s) = \phi(d)$ (as $a+m \leq W$); $\chi(d) = \phi(d)$ (as $d \leq W$); $\chi(E+m) = 1 - \phi(m) = 1 - (1 - \phi(d)) = \phi(d)$ (as $m \leq W$). The triple is monochromatic of colour $\phi(d)$ — contradiction.

(Cb): Suppose $i \in [1, d-1]$ with $\phi(i) \neq \phi(i+1) =: \kappa$, and $m \in [1, d+1]$ with $\phi(m) = \phi(m+s) = \kappa$. Let $y := (d+1-i)a+m$. Then $1 \leq 2a+1 \leq y \leq da+d+1 = E+1 \leq N$. Consider two triples through y :

$\alpha := T(y, i+1) = (y, i+1, E+a+m)$, a solution, since $y+ai+d = (d+1)a+m+d = E+a+m \leq N$ (n

$\beta := T(y, i) = (y, i, E+m)$, a solution, since $y+a(i-1)+d = da+m+d = E+m \leq N$.

Colours of the known cells: $\chi(i) = \phi(i) = 1 - \kappa$ and $\chi(i+1) = \phi(i+1) = \kappa$ (as $i+1 \leq d \leq W$); $\chi(E+a+m) = 1 - \phi(a+m) = \phi(m+s) = \kappa$ (as $a+m \leq W$); $\chi(E+m) = 1 - \phi(m) = 1 - \kappa$ (as $m \leq W$). Now $\chi(y)$ is either κ — making α monochromatic of colour κ — or $1 - \kappa$ — making β monochromatic of colour $1 - \kappa$. Either way a monochromatic solution exists — contradiction. $\square$

Lemma 6.6 (rigidity: ϕ must be s -antiperiodic). *Let $\phi: \mathbb{Z}_{2d} \rightarrow \{0, 1\}$ be antipodal ($\phi(\rho+d) = 1 - \phi(\rho)$) with $\phi(1) = 0$, and suppose that none of the configurations (V), (IV), (Cb) of Lemma 6.5 occurs. Then $\phi(m+s) = 1 - \phi(m)$ for all $m \in \mathbb{Z}_{2d}$; that is, ϕ is s -antiperiodic.*

Proof. First, if $\phi(m^*) = \phi(m^*+s)$ for some $m^* \in \mathbb{Z}_{2d}$ (take the representative $m^* \in [1, 2d]$), then there is an s -plateau with index in $[1, d+1]$: if $m^* \in [1, d+1]$, take $m = m^*$; if $m^* \in [d+2, 2d]$,

take $m = m^* - d \in [2, d]$ — by antipodality $\phi(m) = 1 - \phi(m^*)$ and $\phi(m + s) = 1 - \phi(m^* + s)$, so $\phi(m) = \phi(m + s)$: an in-range s -plateau (of the opposite colour).

Now suppose, for contradiction, that some s -plateau exists; by the previous paragraph the set K of colours of in-range s -plateaus ($m \in [1, d + 1]$) is nonempty. Since (V) does not occur, no in-range plateau has colour $1 - \phi(2)$; hence $K = \{\kappa\}$ with $\kappa := \phi(2)$. Since (IV) does not occur, $1 - \phi(d) \notin K$, so $\phi(d) = \kappa$.

Case $\kappa = 1$: then $\phi(1) = 0 \neq 1 = \phi(2)$, so $i = 1$ (legal, since $d \geq 2$ gives $1 \leq d - 1$) is a transition with $\phi(i + 1) = \kappa$, and an in-range plateau of colour κ exists — configuration (Cb) occurs, a contradiction.

Case $\kappa = 0$: then $\phi(1) = \phi(2) = \phi(d) = 0$. *Sub-claim:* $\phi \equiv 0$ on $[1, d]$. For $d \in \{2, 3\}$ this is immediate — the values $\phi(1) = \phi(2) = \phi(d) = 0$ already exhaust $[1, d]$ — so the branch that follows is vacuous there. Otherwise, let j be minimal in $[1, d]$ with $\phi(j) = 1$; then $3 \leq j \leq d - 1$. Since $\phi(d) = 0$, let i be minimal in $[j, d - 1]$ with $\phi(i + 1) = 0$; by minimality $\phi(i) = 1$ (for $i = j$ directly; for $i > j$, all of $\phi(j), \dots, \phi(i)$ equal 1). Then $i \in [1, d - 1]$, $\phi(i) \neq \phi(i + 1) = 0 = \kappa$, and an in-range plateau of colour κ exists — configuration (Cb) occurs, a contradiction. This proves the sub-claim: $\phi \equiv 0$ on $[1, d]$, and hence, by antipodality, $\phi \equiv 1$ on $[d + 1, 2d]$. But then $m = d + 1 \in [1, d + 1]$ satisfies $\phi(d + 1) = 1$ and $\phi(d + 1 + s) = 1$ (since $d + 2 \leq d + 1 + s \leq 2d$, because $1 \leq s \leq d - 1$): an in-range plateau of colour 1, so $1 \in K = \{0\}$ — a contradiction.

Both cases are impossible, so no s -plateau exists anywhere in $\mathbb{Z}_{2d}$, i.e. $\phi(m + s) = 1 - \phi(m)$ for all m . $\square$

Lemma 6.7 (2-adic obstruction and existence criterion). *Let $d \geq 2$, $q \geq 0$, and $a = 2dq + d + s$.*

(a) *For $s \in [1, d - 1]$, the existence condition $\nu_2(a) \leq \nu_2(a - d)$ is equivalent to $\nu_2(s) \neq \nu_2(d)$. Moreover, the same computation covers $s = d$ (i.e. $r = 2d$): there the existence condition always fails, re-proving Lemma 2.3(i), so that under the existence condition the regime $r \in [d + 1, 2d]$ reduces to $s \in [1, d - 1]$.*

(b) *If $\nu_2(s) \neq \nu_2(d)$, then no $\phi: \mathbb{Z}_{2d} \rightarrow \{0, 1\}$ can be simultaneously antipodal and s -antiperiodic.*

Proof. (a) Write $v := \nu_2(d)$ and $w := \nu_2(s)$, so that $a = 2dq + (d + s)$ and $a - d = 2dq + s$, and note $\nu_2(2dq) \geq v + 1$ whenever $q \geq 1$ (if $q = 0$ the term is absent).

If $w < v$: $\nu_2(d + s) = w$ and $\nu_2(2dq) \geq v + 1 > w$, so $\nu_2(a) = w$; likewise $\nu_2(a - d) = \nu_2(2dq + s) = w$. Hence $\nu_2(a) = \nu_2(a - d)$: the existence condition holds.

If $w > v$: $\nu_2(d + s) = v < \nu_2(2dq)$, so $\nu_2(a) = v$; and $\nu_2(a - d) \geq \min(v + 1, w) \geq v + 1 > \nu_2(a)$: the existence condition holds (strictly).

If $w = v$: write $d = gd'$, $s = gs'$ with $g = \gcd(d, s)$; then $\nu_2(g) = v$ and d', s' are both odd, so $\nu_2(d + s) \geq v + 1$ and hence $\nu_2(a) \geq v + 1$; but $\nu_2(a - d) = \nu_2(2dq + s) = v$ (since $\nu_2(2dq) \geq v + 1 > v = \nu_2(s)$). So $\nu_2(a) > \nu_2(a - d)$: the existence condition fails. This proves the equivalence for $s \in [1, d - 1]$.

Finally, the computation in the case $w = v$ applies verbatim to $s = d$: there $g = d$ and $d' = s' = 1$ are both odd, so again $\nu_2(a) > \nu_2(a - d)$. Thus $r = 2d$ never satisfies the existence condition, and, under the existence condition, the regime $r \in [d + 1, 2d]$ reduces to $s \in [1, d - 1]$.

(b) Suppose ϕ is antipodal and s -antiperiodic: $\phi(x + d) = 1 - \phi(x)$ and $\phi(x + s) = 1 - \phi(x)$ for all $x \in \mathbb{Z}_{2d}$. Composing the two, $\phi(x + (d + s)) = \phi(x)$ and $\phi(x + (d - s)) = \phi(x)$: both $d + s$ and $d - s$ are periods of ϕ . The set P of periods of ϕ is a subgroup of $\mathbb{Z}_{2d}$, so P contains the subgroup generated by $d - s$ and $d + s$, which is generated by $h := \gcd(d - s, d + s, 2d)$. With $g = \gcd(d, s)$ and $d = gd'$, $s = gs'$: $\gcd(d - s, d + s) = g \gcd(d' - s', d' + s')$, and $\gcd(d' - s', d' + s')$

divides both $2d'$ and $2s'$, hence divides $2 \gcd(d', s') = 2$. Since $\nu_2(s) \neq \nu_2(d)$, the integers d' and s' have opposite parities, so $d' + s'$ is odd, so $\gcd(d' - s', d' + s')$ is odd, hence equals 1. Thus $\gcd(d - s, d + s) = g$, and $h = \gcd(g, 2d) = g$ (as $g \mid d$). So $g \in P$, and since $g \mid d$, also $d \in P$: $\phi(x + d) = \phi(x)$ for all x . But antipodality says $\phi(x + d) = 1 - \phi(x)$ — contradiction. $\square$

Theorem 5 (upper bound in the regime $r > d$). *For every $d \geq 2$, $q \geq 0$ and $s \in [1, d - 1]$ with $\nu_2(s) \neq \nu_2(d)$, put $a := 2dq + d + s$, so that $r = d + s > d$ — equivalently, consider any a for which $\text{Rad}_2(a; a - d)$ exists and $r > d$. Then every 2-colouring of $[1, N]$, $N = (a + 3)d + a - (d - 1)$, contains a monochromatic solution of $x_1 + ax_2 - x_3 = a - d$; that is,*

$$\text{Rad}_2(a; a - d) \leq (a + 3)d + a - (d - 1).$$

Proof. Suppose $\chi: [1, N] \rightarrow \{0, 1\}$ has no monochromatic solution; by Lemma 2.2 we may assume $\chi(1) = 0$, and the standing assumption of this section is in force. Lemmas 6.1–6.4 produce the antipodal window pattern ϕ with $\phi(1) = 0$. Since χ is assumed valid, none of the configurations (V), (IV), (Cb) of Lemma 6.5 can occur (each produces an explicit monochromatic solution inside $[1, N]$). By Lemma 6.6, ϕ is s -antiperiodic. By Lemma 6.7(a) and (b), this contradicts the existence condition: $\nu_2(s) \neq \nu_2(d)$ holds, yet no antipodal s -antiperiodic pattern on $\mathbb{Z}_{2d}$ exists. Hence no such χ exists, and $\text{Rad}_2(a; a - d) \leq N$. $\square$

Since $r > d$ gives $\min(r - 1, d - 1) = d - 1$, Theorem 5 is exactly the upper bound required by Theorem 2 in this regime; combined with the extremal family colouring of $[1, N - 1]$ of Section 4, it pins $\text{Rad}_2(a; a - d) = (a + 3)d + a - \min(r - 1, d - 1)$ in the whole regime $r > d$.

Remark 5. All linear range conditions used in the proofs of Lemmas 6.1–6.5 (each cited triple satisfies $x_3 = x_1 + a(x_2 - 1) + d$ with all entries in $[1, N]$) were verified symbolically above and, in addition, mechanically replayed at the boundary values of all free parameters for all 1769 in-scope pairs (a, d) with $d \leq 16$, $a \leq 400$ (including $q = 0$). Likewise, the case analysis of Lemma 6.6 was corroborated by exhaustive enumeration: for every $d \leq 16$ and every s , among all 2^{d-1} antipodal patterns ϕ with $\phi(1) = 0$, those avoiding the three configurations of Lemma 6.5 are exactly the s -antiperiodic ones — and, in accordance with Lemma 6.7, none exists when $\nu_2(s) \neq \nu_2(d)$. See Section 9.

7 The conjectures of Dwivedi and Tripathi

Throughout this section $M := a(a + 3) + 1 = a^2 + 3a + 1$, an odd integer with $M \geq 5$, and for an integer c we set

$$X := 1 + c(a + 3), \quad K := \lceil X/M \rceil - 1.$$

We extend the 2-adic valuation by the conventions $\nu_2(0) := \infty$ and $\nu_2(-n) := \nu_2(n)$ for $n > 0$.

Dwivedi and Tripathi accompany their theorems in [1] with two conjectures for the zones adjacent to them. *Conjecture 1* asserts that

$$\text{Rad}_2(a; c) = (a + 3)(a - c) + 1 \quad \text{for all } a \geq 1 \text{ and } c \leq 0 \text{ with } \nu_2(a) \leq \nu_2(c);$$

they proved this when $a \mid c$ (their Theorem 4) and, for odd a , when c is sufficiently negative, in a precise residue-dependent sense. *Conjecture 2* asserts that

$$\text{Rad}_2(a; c) = K + 1 \quad \text{for all } a \geq 1 \text{ and } c \geq a(K + 2) \text{ with } \nu_2(a) \leq \nu_2(c);$$

their Theorem 6 gives the lower bound $\text{Rad}_2(a; c) \geq K + 1$ for every $c > a$, and their Theorem 7 proves the conjectured equality for all sufficiently large c , without an explicit threshold. In this section we prove both conjectures in full. Conjecture 1 is the case $q = 0$ of the unified upper bound of Section 5 (Theorem 4); Conjecture 2 then follows by a reflection argument, in the spirit of the proof of Theorem 7 in [1], which carries the regime $c \geq a(K + 2)$ onto the regime of Conjecture 1.

Theorem 6 (Conjecture 1 of Dwivedi–Tripathi). *For every $a \geq 1$ and every integer $c \leq 0$ with $\nu_2(a) \leq \nu_2(c)$,*

$$\text{Rad}_2(a; c) = (a + 3)(a - c) + 1.$$

Proof. Set $d := a - c$; since $c \leq 0$, we have $d \geq a \geq 1$.

Lower bound. Theorem 2 of [1] states that $\text{Rad}_2(a; c) \geq (a + 3)(a - c) + 1$ for every integer $c < a$; here $c \leq 0 < 1 \leq a$, so it applies.

Upper bound for $d \geq 2$. We check the hypotheses of Theorem 4 for the pair (a, d) , with the exact bookkeeping of its parameters q and r . Write $a = 2dq + r$ with $q \geq 0$ and $r \in [1, 2d]$; since $1 \leq a \leq d < 2d$, the unique such decomposition is $q = 0$ and $r = a$. The regime condition $r \leq d$ holds because $r = a \leq d$, and the existence condition $\nu_2(a) \leq \nu_2(a - d) = \nu_2(c)$ is our hypothesis (for $c = 0$, i.e. $d = a$, it holds by $\nu_2(0) = \infty$). Theorem 4 therefore gives

$$\text{Rad}_2(a; a - d) \leq (a + 3)d + a - \min(r - 1, d - 1).$$

Since $r = a \leq d$ we have $\min(r - 1, d - 1) = r - 1 = a - 1$, so the bound reads

$$\text{Rad}_2(a; c) \leq (a + 3)d + a - (a - 1) = (a + 3)d + 1 = (a + 3)(a - c) + 1,$$

matching the lower bound.

The excluded pair $d = 1$. If $d = 1$, then $a = c + 1 \leq 1$ forces $(a, c) = (1, 0)$, the unique pair with $c \leq 0$ and $d = 1$. Here $1 \nmid 0$ and $0 \leq 0$, so Theorem 4 of [1] (equality for $a \mid c$, $c \leq 0$) gives $\text{Rad}_2(1; 0) = (1 + 3)(1 - 0) + 1 = 5$ exactly. $\square$

We now turn to Conjecture 2. Fix $a \geq 1$. The proof reflects the interval $[1, K + 1]$ about its midpoint: the involution $\sigma(x) = N + 1 - x$ of $[1, N]$ exchanges solutions of $\mathcal{E}_{a, c}$ with solutions of $\mathcal{E}_{a, c'}$ for the reflected constant $c' = a(N + 1) - c$, and at $N = K + 1$ the hypothesis $c \geq a(K + 2)$ places c' in the range $c' \leq 0$ of Theorem 6. The following lemmas prepare the argument.

Lemma 7.1 (reflection identity). *Fix an integer $N \geq 1$ and let $\sigma: [1, N] \rightarrow [1, N]$, $\sigma(x) = N + 1 - x$. Then σ is an involutive bijection, and for every triple $(x_1, x_2, x_3) \in [1, N]^3$ and every integer c : (x_1, x_2, x_3) solves $\mathcal{E}_{a, c}$ if and only if $(\sigma(x_1), \sigma(x_2), \sigma(x_3))$ solves $\mathcal{E}_{a, \hat{c}}$, where $\hat{c} := a(N + 1) - c$. Moreover, for every 2-colouring χ of $[1, N]$, setting $\chi' := \chi \circ \sigma$, a triple $(y_1, y_2, y_3) \in [1, N]^3$ is χ' -monochromatic if and only if $(\sigma(y_1), \sigma(y_2), \sigma(y_3))$ is χ -monochromatic. Consequently, χ admits a monochromatic solution of $\mathcal{E}_{a, c}$ in $[1, N]^3$ if and only if χ' admits a monochromatic solution of $\mathcal{E}_{a, \hat{c}}$ in $[1, N]^3$. The map $c \mapsto a(N + 1) - c$ is an involution, matching the involutivity of σ .*

Proof. For $x \in [1, N]$ we have $N + 1 - x \in [1, N]$ and $\sigma(\sigma(x)) = N + 1 - (N + 1 - x) = x$, so σ is an involutive bijection of $[1, N]$. For the equation, put $y_i := N + 1 - x_i$; then

$$\begin{aligned} y_1 + ay_2 - y_3 &= (N + 1 - x_1) + a(N + 1 - x_2) - (N + 1 - x_3) \\ &= (N + 1)(1 + a - 1) - (x_1 + ax_2 - x_3) = a(N + 1) - (x_1 + ax_2 - x_3). \end{aligned}$$

Hence $y_1 + ay_2 - y_3 = \hat{c}$ if and only if $x_1 + ax_2 - x_3 = c$. Since σ is an involution, the same identity applied to (y_1, y_2, y_3) gives the converse direction with the involuted constant $a(N+1) - \hat{c} = c$; componentwise application of σ is thus a bijection between the solution set of $\mathcal{E}_{a,c}$ in $[1, N]^3$ and that of $\mathcal{E}_{a,\hat{c}}$ in $[1, N]^3$. For monochromaticity: $\chi'(y_1) = \chi'(y_2) = \chi'(y_3)$ means $\chi(\sigma(y_1)) = \chi(\sigma(y_2)) = \chi(\sigma(y_3))$, which is exactly the χ -monochromaticity of the σ -image triple; the converse follows by involutivity. Combining the two correspondences: if χ has a χ -monochromatic solution (x_1, x_2, x_3) of $\mathcal{E}_{a,c}$, then $(\sigma(x_1), \sigma(x_2), \sigma(x_3))$ solves $\mathcal{E}_{a,\hat{c}}$ and is χ' -monochromatic, because $\chi'(\sigma(x_i)) = \chi(\sigma(\sigma(x_i))) = \chi(x_i)$; and symmetrically. No boundary issues arise: all entries of both triples stay in $[1, N]$ because σ is a bijection of $[1, N]$. $\square$

Lemma 7.2 (exact ceiling arithmetic). *Let $a \geq 1$ and let c be an integer; recall $M = a(a+3)+1$, $X = 1 + c(a+3)$ and $K+1 = \lceil X/M \rceil$.*

(i) *If $c \geq a(K+2)$, then $c > a$.*

For the remaining parts assume $c > a$, and set $s := (K+1)M - X$, $c' := a(K+2) - c$ and $c'' := a(K+1) - c$. Then, with exact integer arithmetic:

(ii) *$K \geq 1$; hence $a(K+2) \geq 3a$, and the hypothesis $c \geq a(K+2)$ forces $c \geq 3a$.*

(iii) *s is an integer with $0 \leq s \leq M-1$.*

(iv) *$KM \leq X-1$, equivalently $KM \leq c(a+3)$.*

(v) *$(a+3)(a-c') + 1 = (K+1) - s$; in particular $(a+3)(a-c') + 1 \leq K+1$, with equality if and only if $M \mid X$.*

(vi) *$(a+3)(a-c'') = X-1-KM+K \geq K$, i.e. $K \leq (a+3)(a-c'')$.*

(vii) *$c'' < a$; this holds for every $c > a$, with no further hypothesis on c .*

(viii) *If $c \geq a(K+2)$, then $c' \leq 0$ and $c'' = c' - a \leq -a < 0$; moreover $a - c' = c - a(K+1) \geq a \geq 1$.*

Proof. Throughout we use the characterisation of the ceiling: for integers p and $q \geq 1$, $\lceil p/q \rceil$ is the unique integer m with $(m-1)q < p \leq mq$.

(i) Suppose $c \geq a(K+2)$. From $K+1 = \lceil X/M \rceil \geq X/M$ and $aX = a + ac(a+3) = a + c(M-1) = cM + (a-c)$ we get, exactly over the rationals,

$$a(K+1) \geq \frac{aX}{M} = c + \frac{a-c}{M}, \quad \text{hence} \quad a(K+2) \geq c + a + \frac{a-c}{M}.$$

If $c \leq a$, the last fraction is nonnegative, so $a(K+2) \geq c + a \geq c + 1 > c$, contradicting $c \geq a(K+2)$. Hence $c > a$. (In particular $K \geq 1$ then holds by (ii), whose proof uses only $c > a$; so a fortiori $K \geq 0$ and $c \geq a(K+2) \geq 2a > a$, with no circular appeal.)

(ii) $c > a \geq 1$ gives $X = 1 + c(a+3) \geq 1 + (a+1)(a+3) = a^2 + 4a + 4 > a^2 + 3a + 1 = M$, so $X \geq M+1$ and $K+1 = \lceil X/M \rceil \geq 2$, i.e. $K \geq 1$. Then $a(K+2) \geq 3a$.

(iii) By the ceiling characterisation, $KM < X \leq (K+1)M$; hence $s = (K+1)M - X \geq 0$ and $s < (K+1)M - KM = M$, so $s \leq M-1$ (s is an integer).

(iv) From the strict left inequality $KM < X$ with both sides integers, $KM \leq X-1 = c(a+3)$.

(v) $a - c' = a - a(K+2) + c = c - a(K+1)$. Compute

$$\begin{aligned} (a+3)(a-c') + 1 &= (a+3)(c - a(K+1)) + 1 = c(a+3) + 1 - a(a+3)(K+1) \\ &= X - (M-1)(K+1) = X - M(K+1) + (K+1) = (K+1) - s. \end{aligned}$$

Since $s \geq 0$, this is $\leq K + 1$, with equality iff $s = 0$ iff $M \mid X$.

(vi) $a - c'' = c - aK$, so $(a + 3)(a - c'') = c(a + 3) - a(a + 3)K = (X - 1) - (M - 1)K = (X - 1 - KM) + K \geq 0 + K = K$, using (iv).

(vii) $c'' < a$ iff $a(K + 1) - c < a$ iff $aK < c$. By (iv), $aKM \leq ac(a + 3) = c(M - 1) = cM - c < cM$, since $c > 0$; dividing the strict integer inequality $aKM < cM$ by $M > 0$ gives $aK < c$.

(viii) $c \geq a(K + 2)$ gives $c' = a(K + 2) - c \leq 0$ and $c'' = c' - a \leq -a < 0$ directly (note $c' - c'' = a(K + 2) - a(K + 1) = a$). Also $a - c' = c - a(K + 1) \geq a(K + 2) - a(K + 1) = a \geq 1$. $\square$

Lemma 7.3 (2-adic transfer). *Let $a \geq 1$ and c be integers with $\nu_2(a) \leq \nu_2(c)$, and let n be any integer. Then $c^* := an - c$ satisfies $\nu_2(c^*) \geq \nu_2(a)$; in particular $\nu_2(a) \leq \nu_2(c^*)$, so the existence condition for $\text{Rad}_2(a; c^*)$ holds. This covers $c^* = 0$, by the convention $\nu_2(0) = \infty \geq \nu_2(a)$.*

Proof. The 2-adic valuation is ultrametric: for integers u and v , $\nu_2(u - v) \geq \min(\nu_2(u), \nu_2(v))$, valid with the conventions above. Indeed, if $u = v$ the left side is ∞ ; otherwise, with $m := \min(\nu_2(u), \nu_2(v))$, $2^m \mid u$ and $2^m \mid v$, hence $2^m \mid (u - v)$, so $\nu_2(u - v) \geq m$. Now $\nu_2(an) = \nu_2(a) + \nu_2(n) \geq \nu_2(a)$ (equal to ∞ if $n = 0$, still $\geq \nu_2(a)$), and $\nu_2(c) \geq \nu_2(a)$ by hypothesis. Therefore $\nu_2(c^*) = \nu_2(an - c) \geq \min(\nu_2(an), \nu_2(c)) \geq \nu_2(a)$. $\square$

Lemma 7.4 (the boundary case $c = a(K + 2)$). *Let $a \geq 1$ and $c > a$, and put $c' := a(K + 2) - c$. Then $c = a(K + 2)$ if and only if $c' = 0$, and in that case the instance of Conjecture 1 attached to the pair (a, c') is $\text{Rad}_2(a; 0) = (a + 3)(a - 0) + 1 = a(a + 3) + 1 = M$, which is Theorem 4 of [1] applied at $c = 0$ (where $a \mid 0$), even independently of Theorem 4. Moreover the boundary family is nonempty for every a : the value $c_0 := a(M + 1)$ satisfies $X = M^2$, $K + 1 = M$ and $c_0 = a(K + 2)$ exactly, with $s = 0$; and $\nu_2(c_0) = \nu_2(a) + \nu_2(M + 1) \geq \nu_2(a)$, so c_0 is admissible for every $a \geq 1$.*

Proof. The first equivalence is immediate from the definition of c' . When $c' = 0$, the pair $(a, 0)$ satisfies $a \mid 0$ and $0 \leq 0$, so Theorem 4 of [1] gives $\text{Rad}_2(a; 0) = (a + 3)(a - 0) + 1 = a(a + 3) + 1 = M$ (existence holds for every a , since $\nu_2(0) = \infty \geq \nu_2(a)$). For the nonemptiness, let $c_0 = a(M + 1)$; using $a(a + 3) = M - 1$,

$$X = 1 + a(M + 1)(a + 3) = 1 + (M + 1)(M - 1) = M^2,$$

so $K + 1 = \lceil M^2/M \rceil = M$, $K + 2 = M + 1$, and $a(K + 2) = a(M + 1) = c_0$ exactly; then $s = (K + 1)M - X = M \cdot M - M^2 = 0$, and by Lemma 7.2(v) the value $(a + 3)(a - c') + 1$ equals $K + 1$: the reflection in the proof of Theorem 7 below is tight there. Finally $\nu_2(c_0) = \nu_2(a) + \nu_2(M + 1) \geq \nu_2(a)$. $\square$

Theorem 7 (Conjecture 2 of Dwivedi–Tripathi). *Let $a \geq 1$ and let c be an integer with $\nu_2(a) \leq \nu_2(c)$ and $c \geq a(K + 2)$, where $K = \lceil (1 + c(a + 3))/(1 + a(a + 3)) \rceil - 1$. Then*

$$\text{Rad}_2(a; c) = K + 1.$$

Before the proof, one remark on the hypotheses: any such c automatically satisfies $c > a$. Indeed, by Lemma 7.2(i), the inequality $c \geq a(K + 2)$ alone forces $c > a$, whence $K \geq 1$ by Lemma 7.2(ii) and in fact $c \geq a(K + 2) \geq 3a$. The standing regime $c > a$ of Theorems 6 and 7 of [1] is therefore implied, not additionally assumed.

Proof. Write $N := K + 1$; as just noted, $c > a$ and $K \geq 1$, so $N \geq 2$.

Lower bound. Theorem 6 of [1] states that $\text{Rad}_2(a; c) \geq \lceil (1 + c(a + 3)) / (1 + a(a + 3)) \rceil = K + 1$ for every $c > a$, which holds here. (Remark 6 gives an alternative derivation from Theorem 2 of [1] by reflection.)

Upper bound. Let χ be an arbitrary 2-colouring of $[1, N]$. Set $\sigma(x) := N + 1 - x = K + 2 - x$ on $[1, N]$, $\chi' := \chi \circ \sigma$, and $c' := a(N + 1) - c = a(K + 2) - c$.

Step 1: Theorem 6 applies to the pair (a, c') . By hypothesis $c \geq a(K + 2)$, so $c' \leq 0$ (Lemma 7.2(viii)). By Lemma 7.3 with $n = K + 2$, $\nu_2(c') \geq \nu_2(a)$, i.e. $\nu_2(a) \leq \nu_2(c')$. Hence Theorem 6 yields

$$\text{Rad}_2(a; c') = (a + 3)(a - c') + 1 =: R'.$$

(In the boundary case $c = a(K + 2)$ one has $c' = 0$ and $R' = M$; see Lemma 7.4.)

Step 2: R' fits inside $[1, N]$. By Lemma 7.2(v), $R' = (K + 1) - s$ with $s = (K + 1)M - X \in [0, M - 1]$ (Lemma 7.2(iii)), hence $R' \leq K + 1 = N$. Also $R' \geq M \geq 5$: indeed $a - c' \geq a$ by Lemma 7.2(viii), so $R' = (a + 3)(a - c') + 1 \geq (a + 3)a + 1 = M$.

Step 3: a monochromatic c' -solution for χ' . The map χ' is a 2-colouring of $[1, N]$; restrict it to $[1, R']$. Since $R' = \text{Rad}_2(a; c')$, every 2-colouring of $[1, R']$ — in particular $\chi'|_{[1, R']}$ — admits a monochromatic solution $(y_1, y_2, y_3) \in [1, R']^3$ of $\mathcal{E}_{a, c'}$. As $R' \leq N$, this triple lies in $[1, N]^3$ and is χ' -monochromatic.

Step 4: reflect back. By Lemma 7.1, applied with this N , the triple $(x_1, x_2, x_3) := (\sigma(y_1), \sigma(y_2), \sigma(y_3))$ lies in $[1, N]^3$, solves $\mathcal{E}_{a, \hat{c}}$ with $\hat{c} = a(N + 1) - c' = a(K + 2) - (a(K + 2) - c) = c$, and is χ -monochromatic, since $\chi' \circ \sigma = \chi \circ \sigma \circ \sigma = \chi$ by involutivity.

Since χ was arbitrary, every 2-colouring of $[1, K + 1]$ admits a monochromatic solution of $\mathcal{E}_{a, c}$ in $[1, K + 1]^3$; in particular $\text{Rad}_2(a; c)$ is finite and $\text{Rad}_2(a; c) \leq K + 1$. Combined with the lower bound, $\text{Rad}_2(a; c) = K + 1$. $\square$

Remark 6. The lower bound in Theorem 7 can also be re-derived from Theorem 2 of [1] alone, without the construction underlying their Theorem 6, by the same reflection. Let $c > a$ and put $c'' := a(K + 1) - c$; then $c'' < a$ by Lemma 7.2(vii). The proof of Theorem 2 of [1] exhibits, for every integer $c'' < a$ and with no 2-adic hypothesis, an explicit valid colouring χ_0 of $[1, L]$, $L := (a + 3)(a - c'')$, for the pair (a, c'') . This reading matters when $\nu_2(a) > \nu_2(c'')$: then $\text{Rad}_2(a; c'')$ is infinite, and the inequality “ $\text{Rad}_2(a; c'') \geq L + 1$ ” carries its content through the explicit construction — the colouring exists by construction, not by formal citation of an inequality between possibly infinite quantities (cf. Lemma 2.1). By Lemma 7.2(vi), $K \leq L$, so the restriction $\chi_0|_{[1, K]}$ is a valid colouring of $[1, K]$ for (a, c'') (Lemma 2.1). Now apply Lemma 7.1 with $N = K$: the colouring $\chi := \chi_0|_{[1, K]} \circ \sigma_K$, where $\sigma_K(x) = K + 1 - x$, admits a monochromatic solution of $\mathcal{E}_{a, \hat{c}}$ with $\hat{c} = a(K + 1) - c'' = c$ if and only if $\chi_0|_{[1, K]}$ admits one of $\mathcal{E}_{a, c''}$ — which it does not. Hence χ is a valid colouring of $[1, K]$ for (a, c) , and $\text{Rad}_2(a; c) \geq K + 1$ for every $c > a$.

Remark 7. The hypothesis $c \geq a(K + 2)$ holds for every $c \geq 2aM = 2a(a^2 + 3a + 1)$. Indeed $K + 1 = \lceil X/M \rceil \leq (X + M - 1)/M$, so, exactly over the rationals,

$$a(K + 2) \leq \frac{a(X + M - 1)}{M} + a = \frac{c(M - 1) + aM}{M} + a = c - \frac{c}{M} + 2a,$$

which is $\leq c$ as soon as $c \geq 2aM$. Theorem 7 therefore applies to every admissible $c \geq 2a(a^2 + 3a + 1)$; in particular, “ c sufficiently large” in Theorem 7 of [1] may be taken to mean $c \geq 2a(a^2 + 3a + 1)$. Below this threshold the hypothesis is a genuine restriction: for $a = 5$

(where $M = 41$), every multiple of 5 in $[210, 410]$ satisfies $c = a(K + 2)$ exactly — the first being $c_0 = a(M + 1) = 210$ of Lemma 7.4, with the slack s sweeping $[0, M - 1]$ along the family — while $c = 211$ fails it (there $a(K + 2) = 215 > c$). In the remaining zone $a < c < a(K + 2)$, Theorem 6 of [1] still gives the lower bound $\text{Rad}_2(a; c) \geq K + 1$, but equality can fail; this is exactly the middle band studied in Section 8, outside the scope of Conjecture 2.

8 The middle band

Theorems 1, 2, 6 and 7 determine $\text{Rad}_2(a; c)$ everywhere except in the *middle band* $a < c < a(K + 2)$ (with K as in Theorem 7), where the behaviour is qualitatively different from that in all the other zones. About the band, the status of the statements below is as follows. *Theorem 8 is proved here in full. Theorems 9 and 10 are announced:* we state them precisely and describe the structure of their proofs, but the complete arguments are longer and will appear in the companion manuscript [5]; all supporting computations and verification programs are already available with this note. Nothing in Sections 2–7 depends on Theorems 9 or 10.

The multiples of a inside the band are governed by a simple degeneracy, which extends Theorem 5 of [1] (proved there for $1 < m \leq a + 1$) to the full range in which the value m is attained.

Theorem 8. *Let $a \geq 1$ and let $c = ma$ with $2 \leq m \leq a(a + 3) + 1$. Then $\text{Rad}_2(a; ma) = m$.*

Proof. The triple $(x_1, x_2, x_3) = (m, m, m)$ satisfies $m + am - m = am = c$ and is monochromatic under every colouring, so $\text{Rad}_2(a; ma) \leq m$. For the lower bound, Theorem 6 of [1] gives $\text{Rad}_2(a; c) \geq K + 1$, and with $M := a(a + 3) + 1$ one computes $1 + ma(a + 3) = mM - (m - 1)$, whence $K + 1 = \lceil (mM - (m - 1))/M \rceil = m$ for $m \leq M$. $\square$

For c deep inside the band the value is pinned to the immediate neighbourhood of the lower bound of [1], and the exact threshold turns out to be — remarkably — a Rado number from the window $0 < c < a$, so that the band inherits its fine structure from Theorems 1 and 2.

Theorem 9 (deep tail). *Let (a, c) lie in the middle band with $\nu_2(a) \leq \nu_2(c)$ and $a \nmid c$, write $t = c \bmod a \in [1, a - 1]$ and $\lambda = \lceil c/a \rceil$, and set*

$$E_1 = E_1(a, t) := \text{Rad}_2(a; a - t),$$

the value determined by Theorem 1 (for $t = 1$) and Theorem 2 (for $t \geq 2$); this Rado number exists, since $\nu_2(a) \leq \nu_2(c)$ forces $\nu_2(a) \leq \nu_2(t)$, hence $\nu_2(a) \leq \nu_2(a - t)$. If $t \geq 2$ and $\lambda \geq t(a + 3) + 2$, or $t = 1$ and $\lambda \geq 2a - 2$, then

$$\text{Rad}_2(a; c) = \lambda - 1 + [\lambda \leq E_1],$$

where $[P]$ equals 1 if the condition P holds and 0 otherwise.

In the remaining (shallow) part of the band no closed formula appears to hold, and the computations indicate that this is not an artefact of our methods. Along each diagonal $\{c : c \equiv -\mu \pmod{a}\}$ the excess $\text{Rad}_2(a; c) - (K + 1)$ forms a non-increasing staircase whose step pattern follows the three-distance (Sturmian) structure of the rotation by $c - a$ on $\mathbb{Z}_a$, with step sizes governed by the continued-fraction expansion of μ/a ; the staircases are eventually periodic only for special μ . A concrete finite obstruction: for $(a, c) = (7, 16)$ every valid colouring of the

extremal interval $[1, 15]$ has at least nine maximal runs (an exhaustive check), so no family of colourings with a bounded number of blocks can witness the lower bounds uniformly across the band. The exact values can nonetheless be determined:

Theorem 10 (exact determination of the band). *There is an algorithm which, given (a, c) in the middle band with $\nu_2(a) \leq \nu_2(c)$, computes $\text{Rad}_2(a; c)$ exactly in time polynomial in a .*

The algorithm decides, for each candidate length N , the existence of a valid colouring of $[1, N]$ as follows. For $a \nmid c$ (the multiples of a being settled by Theorem 8), all solution triples of (1) share the multiplier window $W = \{k \in [1, N] : 1 \leq |ak - c| \leq N - 1\}$, a set of $O(N/a)$ consecutive integers; once the colours of W are fixed, validity is a 2-SAT instance; and it suffices to test the $O(|W|^2)$ assignments in which one colour class of W is a single contiguous block. The completeness of this restricted family — every colourable length admits a valid colouring of the restricted form — is the main technical result behind the theorem: in any valid colouring the two central multipliers of W must receive the same colour, and every non-extremal multiplier inherits that colour, by a rigidity argument whose only arithmetic input is, once again, the existence condition $\nu_2(a) \leq \nu_2(c)$.

Across the 28,000-plus band values computed to date the algorithm agrees with the SAT-computed ground truth without exception, and each value carries the same two-sided certificates as the rest of this note.

Example 11. Take $(a, c) = (7, 16)$, so $t = 2$, $\lambda = 3$, $\mu = 5$ and $K + 1 = 3$, while in fact $\text{Rad}_2(7; 16) = 16$ (a certified value; see Section 9): the lower bound of [1] is off by 13 here. An extremal colouring of $[1, 15]$ is

$$1\ 0\ 0\ 1\ 1\ 0\ 1\ 1\ 1\ 0\ 1\ 1\ 0\ 0\ 1,$$

which has nine maximal runs — and, as noted above, no valid colouring of $[1, 15]$ has fewer. At $N = 15$ the multiplier window is $W = \{1, 2, 3, 4\}$, with shifts $7k - 16$ equal to $-9, -2, 5, 12$; the two central multipliers $k = 2, 3$ carry the smallest distances $t = 2$ and $\mu = 5$. The restriction of the colouring above to W is $(1, 0, 0, 1)$ — just three runs, exactly the shape required by the restricted family of Theorem 10. The colouring is globally complex, but its window trace is simple; this contrast is the heart of the band.

9 Computations and certificates

The exact values of $\text{Rad}_2(a; c)$ were computed for all existing pairs with $3 \leq a \leq 15$ and $-100 \leq c \leq 300$ (3662 pairs), and additionally for all existing pairs with $0 < c < a$ up to $a = 45$. The encoding is the obvious one: a Boolean variable per integer in $[1, N]$ and, for every solution triple (1), two clauses forbidding the two monochromatic colourings of that triple. For each pair, the exact value $\text{Rad}_2(a; c) = N$ is certified by:

1. a valid colouring of $[1, N - 1]$, re-verified by an independent brute-force checker of about 30 lines that has no code in common with the search program; and
2. a DRAT proof of unsatisfiability of the instance at $[1, N]$, produced by CaDiCaL 2.1.3 and verified with `drat-trim` [6] against a CNF regenerated by a second, independently written encoder.

As validation, the same programs reproduce all 42 values proved in [1] that fall in the computed range, as well as every value in the ranges covered by Theorems 1 and 2.

Beyond the window $0 < c < a$, the computations corroborate Theorems 6 and 7 at every computed point: all 925 values with $c \leq 0$ in the grid (including the 234 cases that were open before this note) match $(a + 3)(a - c) + 1$, as do 8 further out-of-sample values with a up to 33 computed after the fact; and all 300 computed values with $c \geq a(K + 2)$ match $K + 1$ (12 of them lie outside the ranges covered by the partial results of [1]).

In the middle band the computations yield over 28,000 further exact values (including complete sweeps of the band for several values of a), against which the results of Section 8 were checked without exception.

All witnesses, DRAT certificates, the two independent encoders, the checker, and scripts reproducing every finite verification used in the proofs are available from the author and in the repository accompanying this note, <https://github.com/4Gatitos/RadoNumbers>.

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